

NomLoad

COUNTER

User Manual

Truck loading counter for bags and cartons

Version 1.0.0

Devnom

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1. Introduction

NomLoad Counter counts bags and cartons as they are loaded onto a truck, using the CCTV cameras already installed at the loading bay. It reads the truck's number plate and compares the counted quantity with the gate pass or invoice. If the quantity does not match, the truck is held and an alarm sounds until a supervisor releases it with a PIN and a written reason.

Every loading is recorded with a video clip, a photo of each counted item and the plate image. These files are protected with a digital fingerprint (SHA-256) so any later change can be detected.

1.1 How it works

1. The clerk selects the gate pass for the truck on the **Live Bays** screen.
2. The camera watches the truck entrance. Each bag or carton that crosses the counting line **into** the truck adds 1; an item carried back **out** subtracts 1.
3. The number plate is read automatically and matched with the gate pass.
4. The clerk presses **Close & verify**. The result is **CLEARED** or **MISMATCH**.
5. Every truck can be searched later, and a certificate or evidence pack can be printed.

1.2 User roles

Role	What the user can do
Clerk	Run loading sessions, enter and import gate passes, correct plates, view history and reports. Cannot delete or change counted records.
Supervisor	Everything a clerk can do, plus: release mismatched trucks with PIN, review doubtful counts, adjust counts with a reason, edit products.
Admin	Everything, including cameras, AI models, users, settings and licence.

2. Installation and first start

2.1 Computer requirements

Item	Minimum	Recommended
Windows	Windows 10 64-bit	Windows 10/11 64-bit
Processor	Intel Core i5 (6th gen) or equal	Core i5/i7 10th gen or newer
Memory	8 GB	16 GB
Disk	20 GB free	500 GB+ for long evidence history
Network	Same network as the cameras / NVR	Wired Gigabit
Bays per PC (CPU only)	2	4 or more with a GPU

2.2 Installing

1. Run **NomLoadCounter_Setup_1.0.0.exe** and follow the setup wizard.
2. Tick *Create a desktop shortcut*. On the gate PC also tick *Start NomLoad when Windows starts*.
3. Start NomLoad from the desktop icon. The start screen appears while the AI models load.



Figure 1: Start screen

2.3 Logging in

Enter your username and password and press **Log in**. After installation the first account is **admin** with password **admin123** and supervisor PIN **1234**. You will be asked to choose a new password straight away.

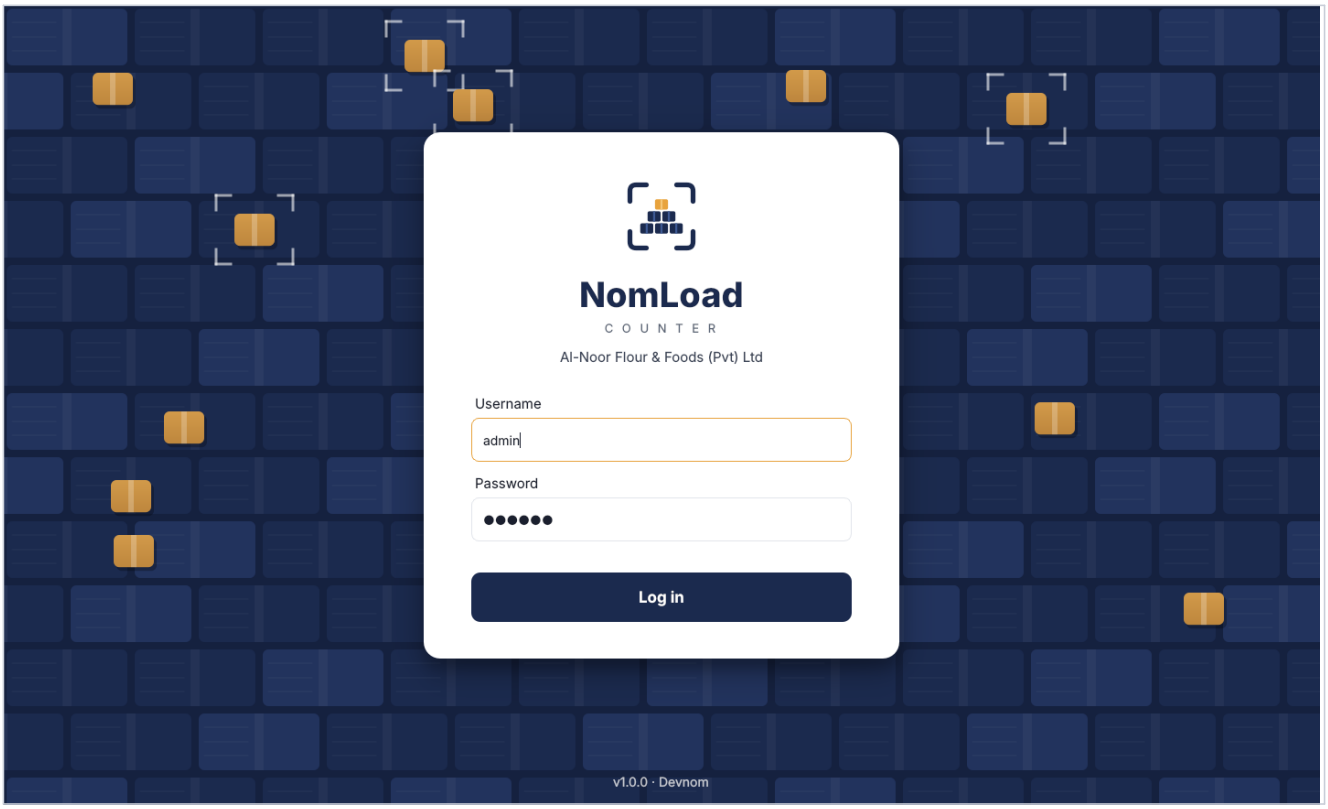


Figure 2: Login screen

Five wrong passwords lock the account for two minutes. An Admin can reset a forgotten password under **Users**.

2.4 Activating the licence

1. Log in as Admin and open **Settings**. Scroll to **Licence**.
2. Paste the licence key (it starts with **NLC-**) and press **Activate**.
3. The status bar at the bottom-left changes to *Professional: Licensed* (or your tier).

Without a key NomLoad runs as a 14-day trial with 2 bays, and reports carry a TRIAL watermark. A key locks to the first PC it is activated on. If the PC is replaced, send the **PC request code** shown in the same section to Devnom for a new activation key.

3. Main screen

The menu on the left opens each part of the program. Users only see the pages their role allows. At the bottom-left are the user's name and role and links for **Appearance** (light / dark), **Change password** and **Log out**.

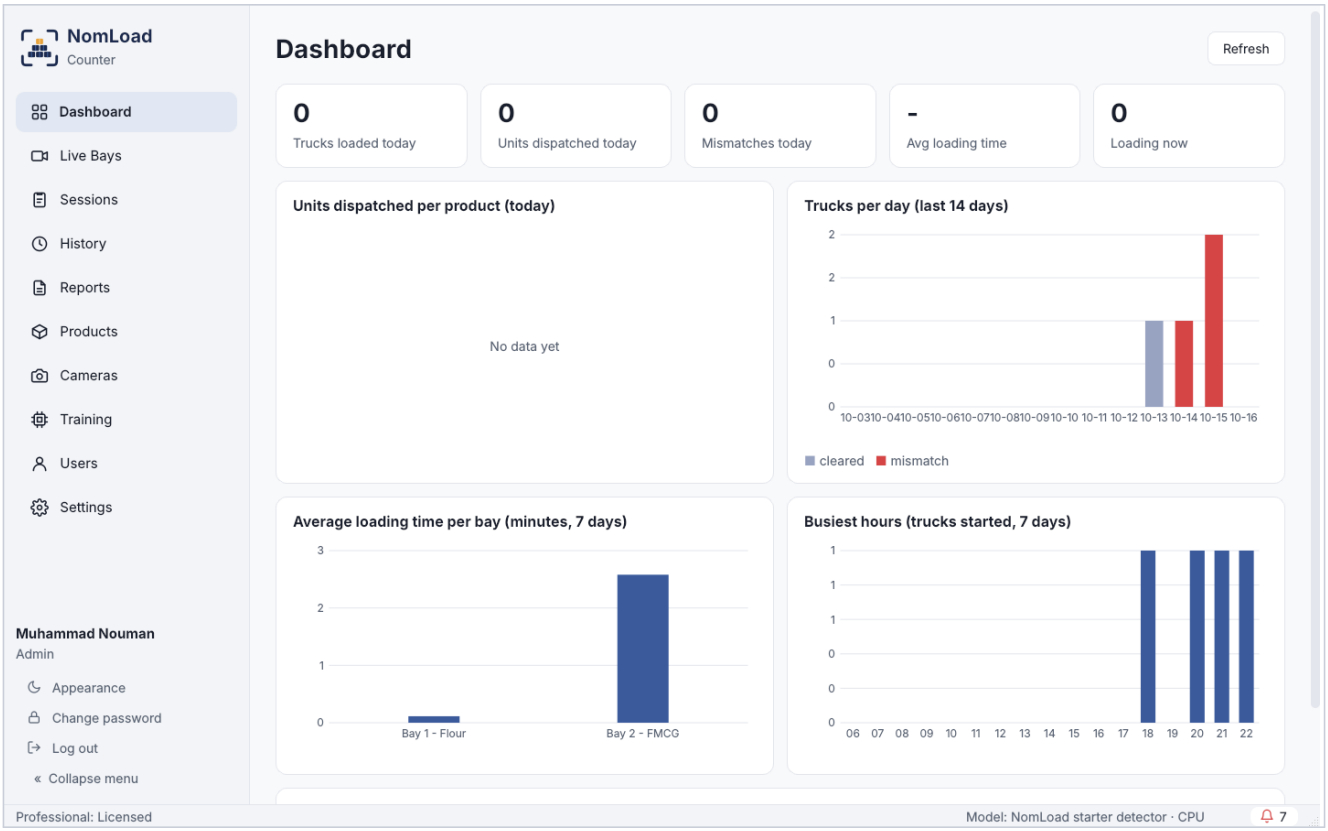


Figure 3: Dashboard with the menu on the left

Menu	Use it to
Dashboard	See today's trucks, units dispatched, mismatches, loading times and busiest hours.
Live Bays	Watch the cameras and run loading sessions.
Sessions	Manage gate passes, today's trucks and the review queue.
History	Search past trucks, view evidence, print certificates.
Reports	Daily dispatch, mismatch report, Excel export, audit log.
Products	Product list, units per stack and tolerance.
Cameras	Bays, cameras and calibration (Admin).
Training	Teach the AI your own bags and cartons (Admin).
Users	User accounts and roles (Admin).
Settings	Company details, alerts, performance and licence (Admin).

The **status bar** at the bottom shows the licence, the AI model in use, messages waiting to be sent and the alert bell. Click the bell to see all alerts.

On smaller screens (for example 1366 × 768) the menu shrinks to icons. Click **»** at the bottom of the menu to open it again.

4. Setting up bays and cameras

A **bay** is one truck loading point. Each bay needs one **counting camera** looking at the truck entrance. A **plate camera** is optional; without it the plate is read from the counting camera.

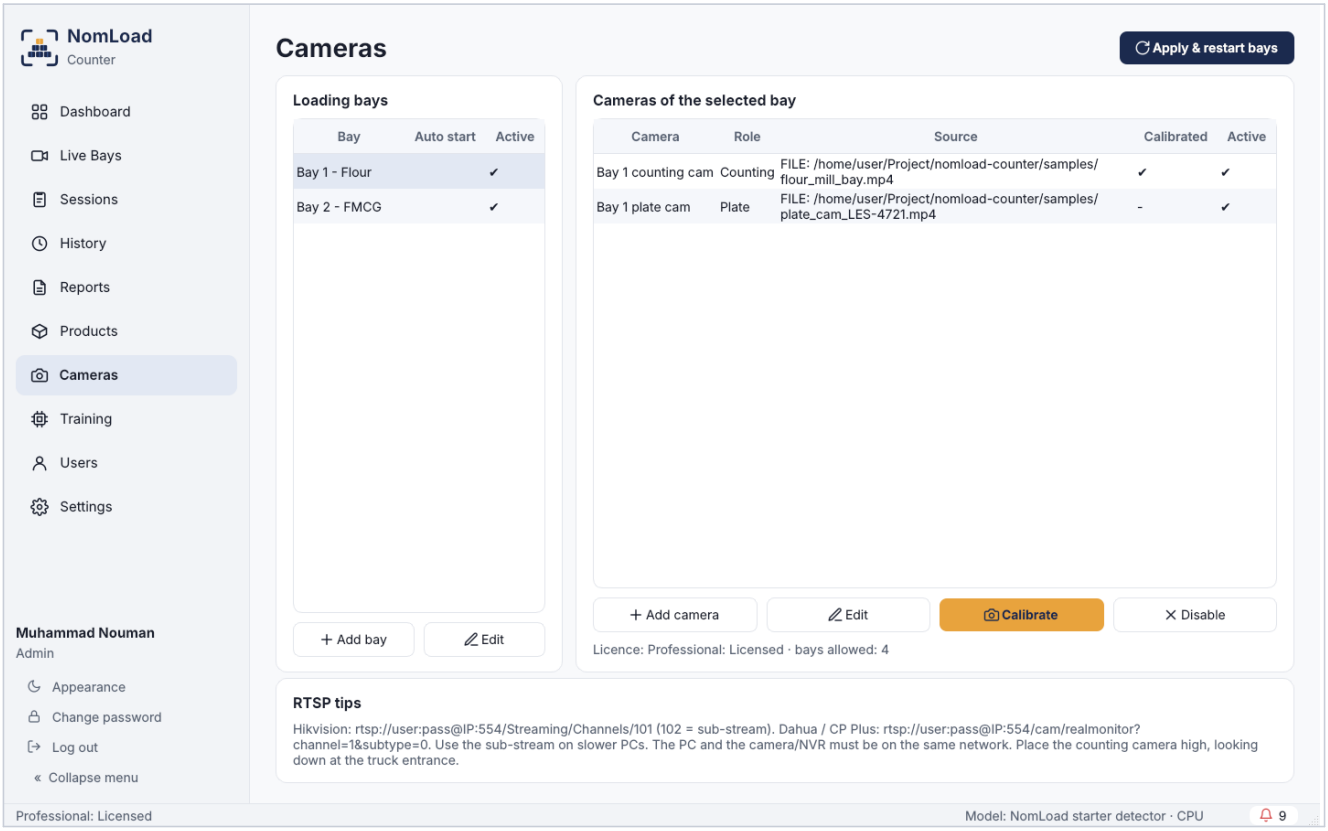


Figure 4: Cameras page: bays on the left, cameras of the selected bay on the right

4.1 Adding a bay

1. Open **Cameras** and press **Add bay**.
2. Enter a name such as *Bay 1 - Flour*.
3. Tick *Start a session automatically when a truck is detected* if trucks should be counted even before the clerk selects a gate pass.

4.2 Adding an IP camera (Hikvision, Dahua and others)

1. Select the bay and press **Add camera**.
2. Choose the role: **Counting camera** or **Number plate camera**.
3. Source: **IP camera / NVR (RTSP)**. Select the brand and enter the IP address, username, password and channel. The RTSP address is filled in automatically.
4. Tick **Use sub-stream** on ordinary PCs. It is enough for counting and uses far less processing power.
5. Press **Test connection**. A picture from the camera must appear.
6. Press **Save**.

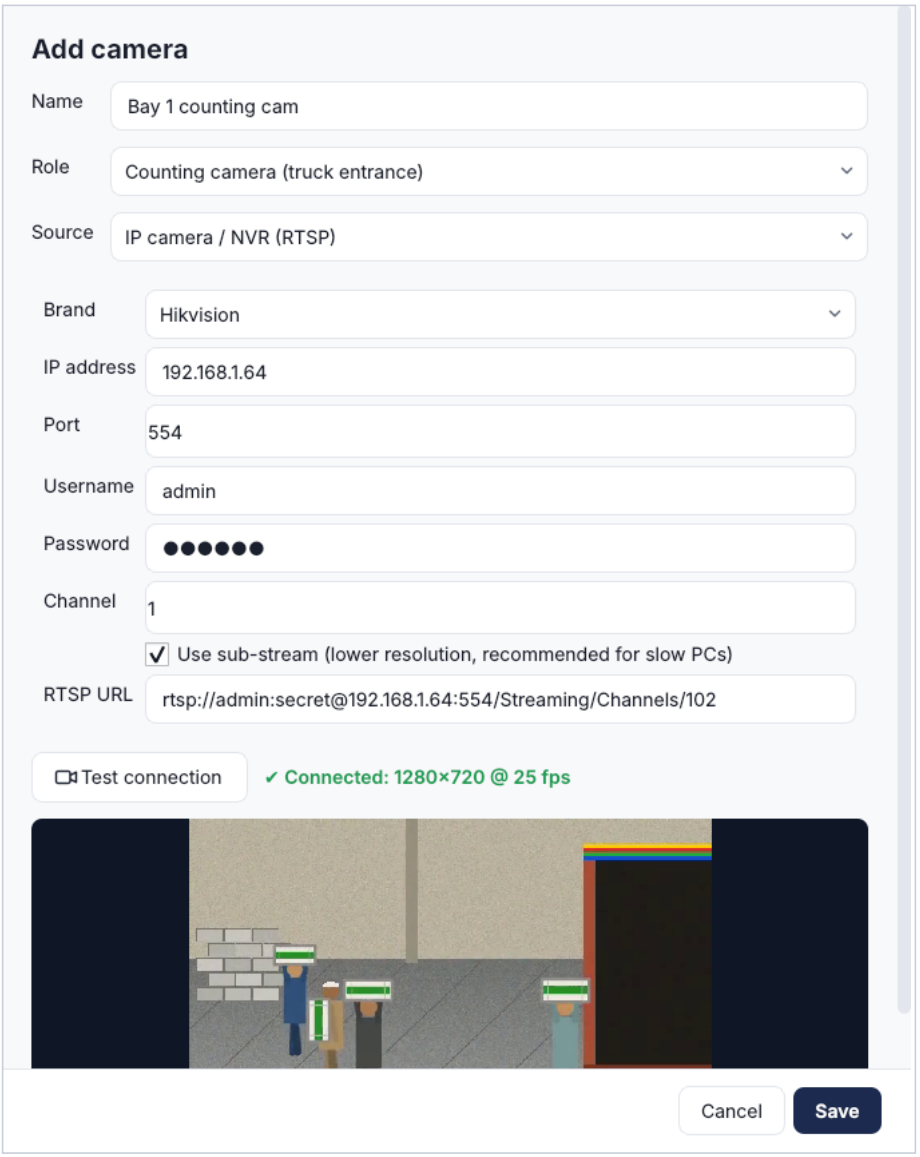


Figure 5: Add camera with the RTSP helper and a successful connection test

Brand	Address format
Hikvision / HiLook	rtsp://user:pass@IP:554/Streaming/Channels/101 (102 = sub-stream)
Dahua / CP Plus	rtsp://user:pass@IP:554/cam/realmonitor?channel=1&subtype=0 (1 = sub-stream)
Uniview	rtsp://user:pass@IP:554/unicast/c1/s0/live
USB webcam	Source: USB webcam, device 0
Recorded video	Source: Video file (for testing and training)

4.3 Calibrating the counting line

Calibration tells NomLoad where the truck entrance is. Do it once per camera, and again if the camera is moved.

1. Select the camera and press **Calibrate**. A live picture opens.

2. Choose **Counting line** and drag a line across the truck entrance, from one side of the tailgate to the other.
3. Check the amber arrow. It must point **into the truck**. If not, press **Flip direction**.
4. Optional: draw **Ignore zones** over areas that must never be counted, such as the stock pile or a walkway. Click the corners and double-click to finish.
5. For the plate, choose **Plate region** and drag a box where the number plate appears.
6. Keep **Save this view as the tamper-detection reference** ticked and press **Save calibration**.
7. Press **Apply & restart bays** on the Cameras page.

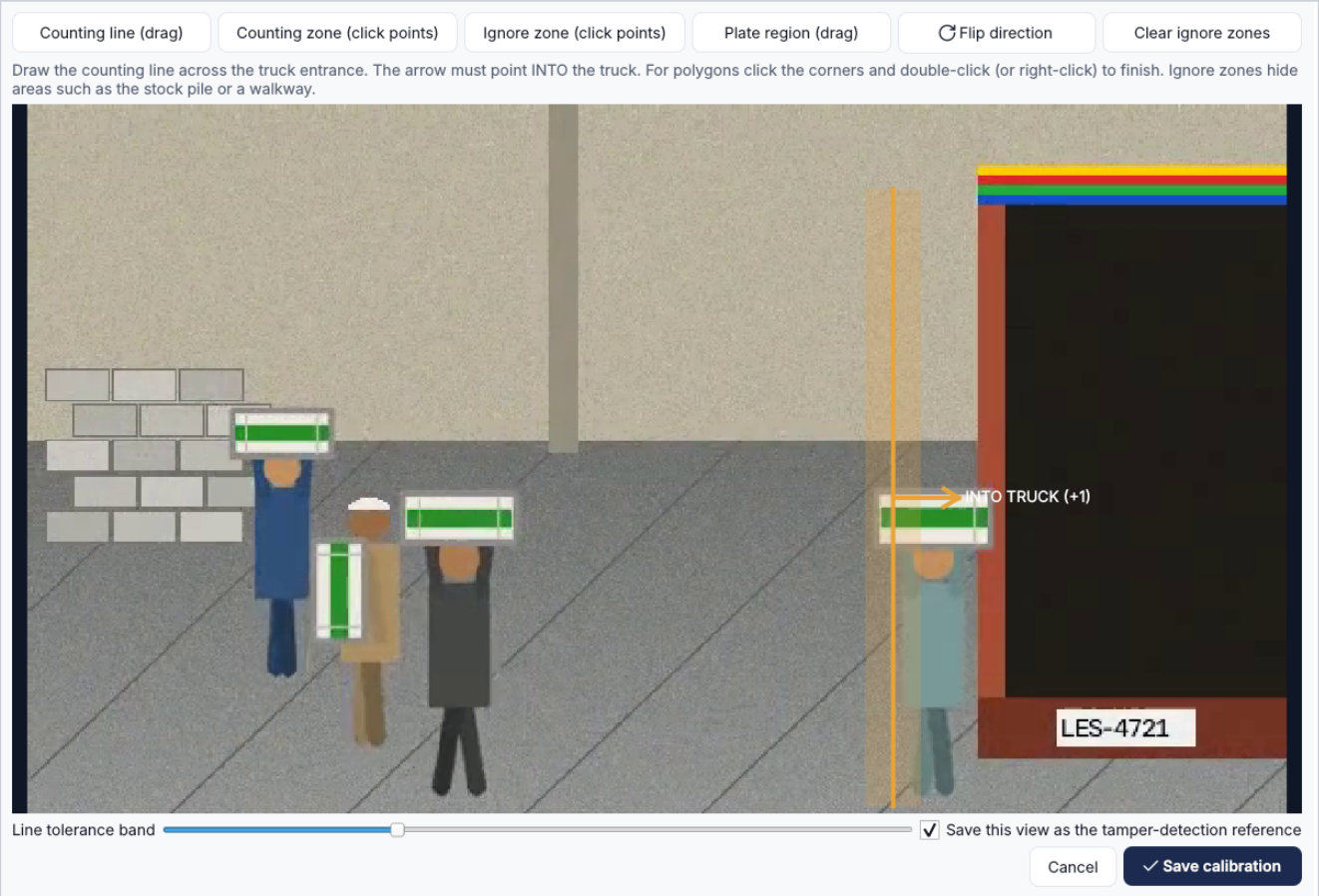


Figure 6: Calibration: counting line with direction arrow and tolerance band

The shaded band around the line prevents double counting when a labourer stops or sways on the line. Widen it with **Line tolerance band** if workers often stand on the line.

5. Products

Every item on a gate pass must exist as a product. The product tells NomLoad what the camera should look for and how many units each detected item is worth.

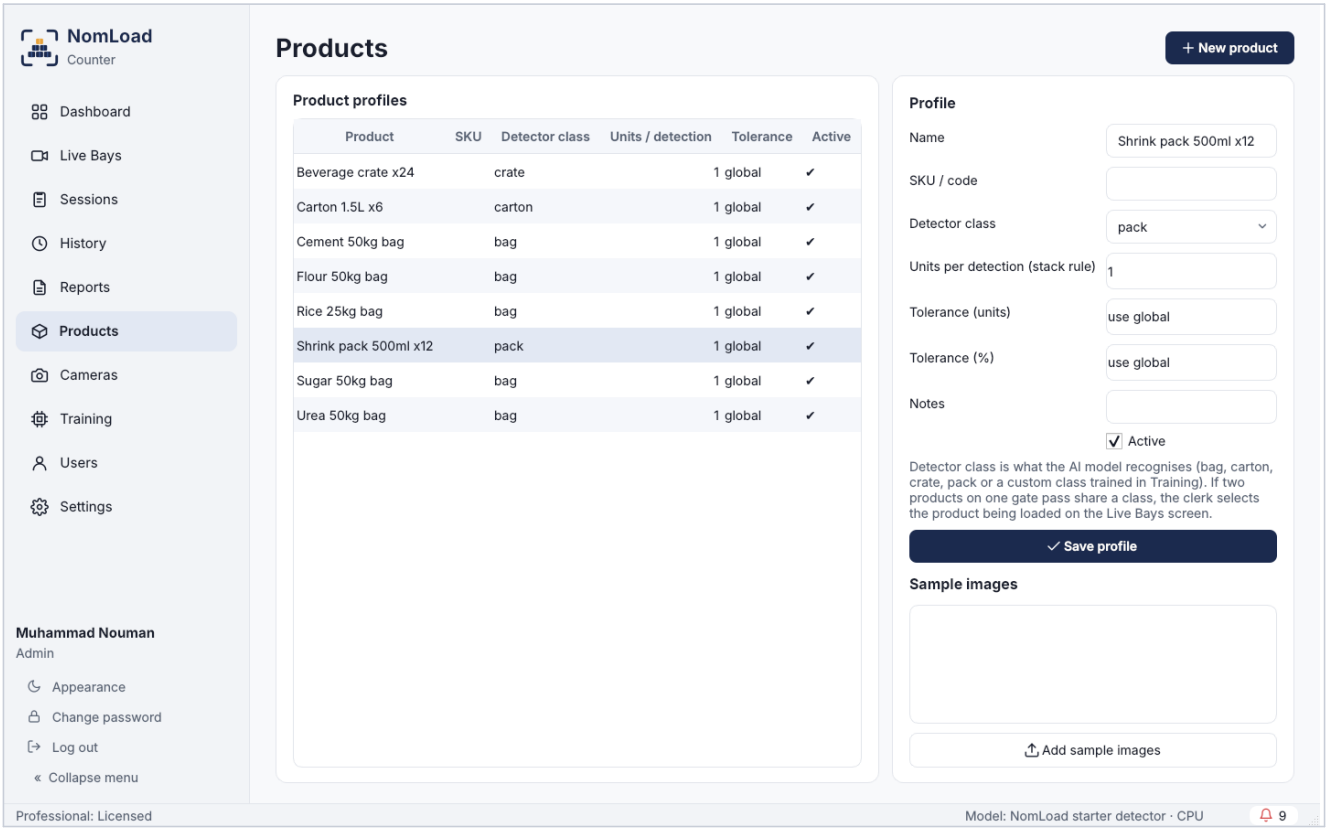


Figure 7: Products page

Field	Meaning
Name / SKU	As written on your gate passes and invoices.
Detector class	What the camera recognises: bag, carton, crate, pack, or a custom class trained for your site.
Units per detection	Units counted for each detected item. Example: a shrink pack of 12 bottles counted as 12 units, or 1 if the gate pass counts packs.
Tolerance	Allowed difference for this product. Empty = the general tolerance from Settings.
Sample images	Photos of the product for reference and training.

If one gate pass has two products of the same class (for example flour bags and sugar bags), the clerk selects the product being loaded on the Live Bays screen. Counts go to the selected product.

6. Gate passes

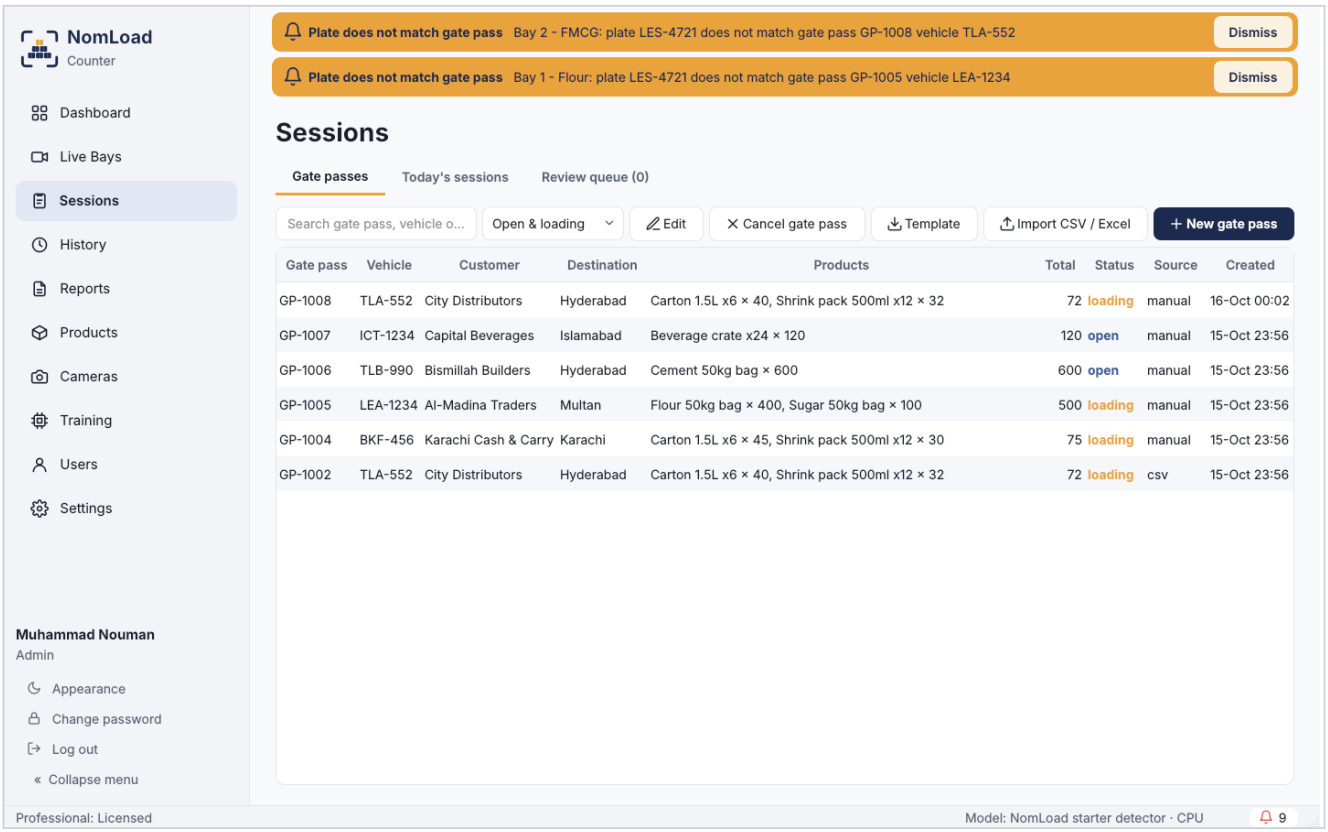


Figure 8: Sessions → Gate passes

6.1 Entering a gate pass

1. Open **Sessions** and press **New gate pass**.
2. Enter the gate pass or invoice number, vehicle number, customer, driver and destination.
3. Select each product and type the quantity. Use **Add product** for more lines.
4. Press **Save**.

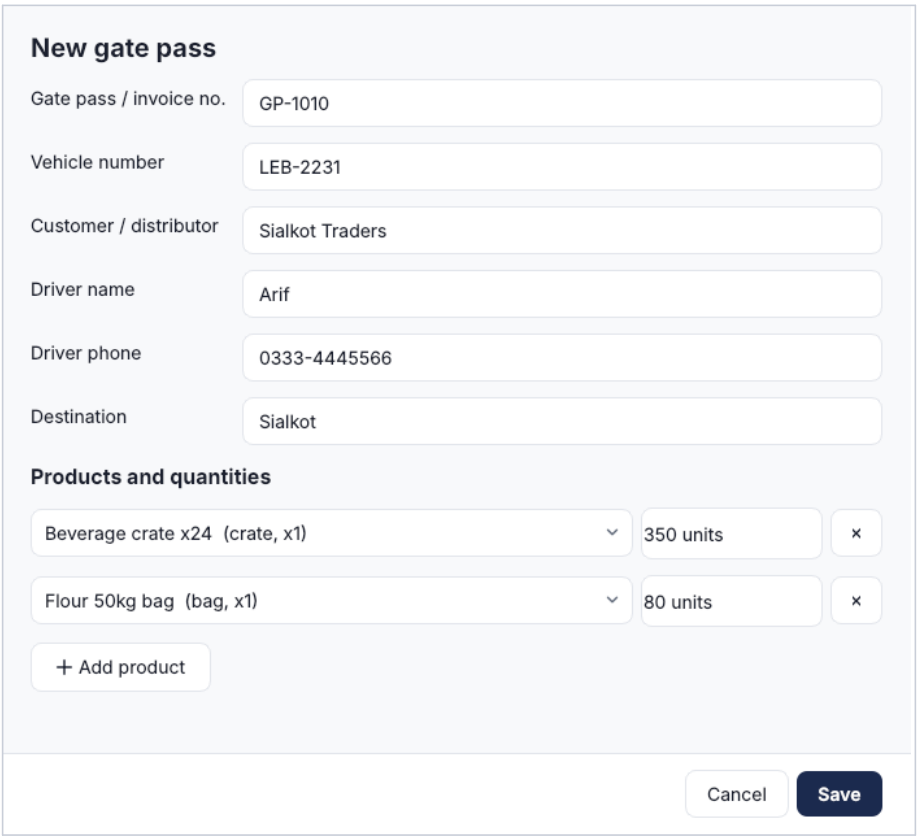


Figure 9: New gate pass with two products

6.2 Importing from Excel or CSV

Press **Template** to save an example file, fill it in Excel, then press **Import CSV / Excel**. One row per product; rows with the same gate pass number become one gate pass.

gate_pass	vehicle	customer	product	quantity
GP-1001	LES-4721	Al-Madina Traders	Flour 50kg bag	400
GP-1001	LES-4721	Al-Madina Traders	Sugar 50kg bag	100
GP-1002	TLA-552	City Distributors	Carton 1.5L x6	600

Optional columns: driver, phone, destination. Product names must match the Products page.

7. Loading a truck

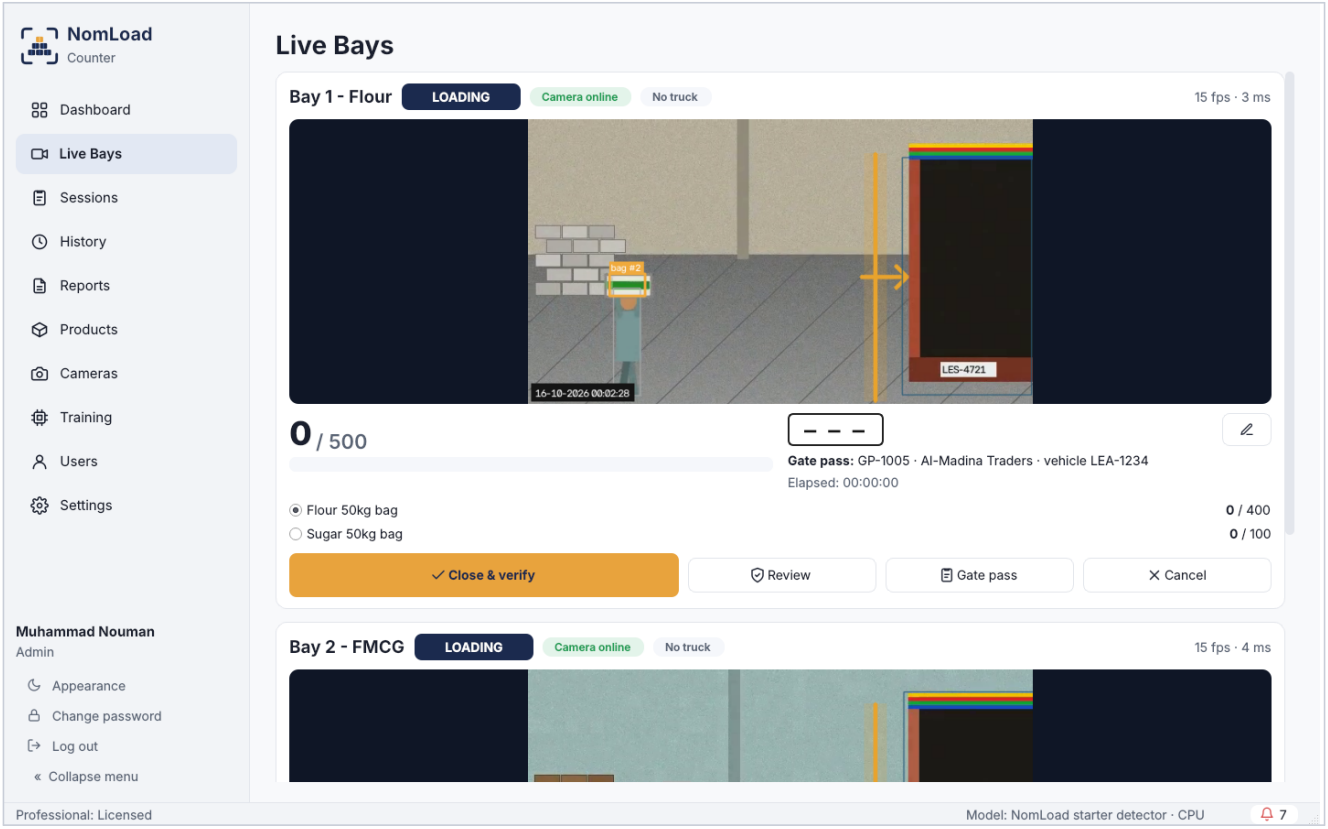


Figure 10: Live Bays with two bays loading

7.1 Starting

1. Open **Live Bays**. Check that the camera shows *Camera online*.
2. When the truck is in position press **Start loading**.
3. Select the truck's gate pass (search by number, vehicle or customer) and press **Start loading**. If the gate pass is not in the list, press **New gate pass**.

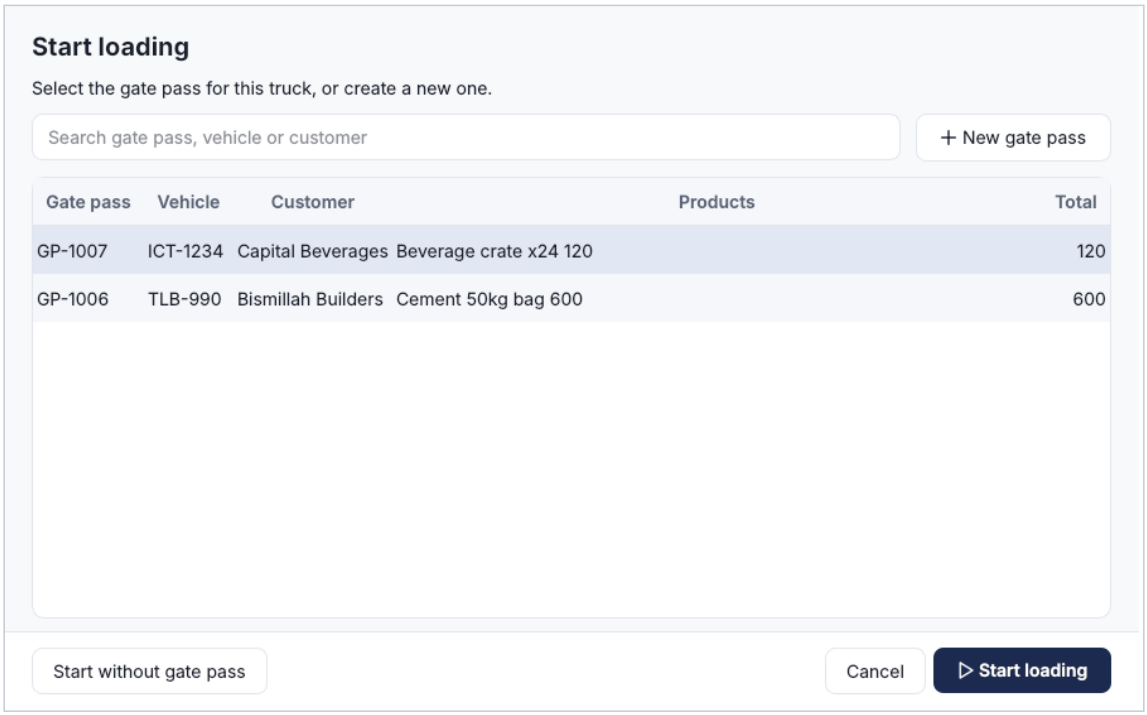


Figure 11: Selecting the gate pass

7.2 During loading

Each counted item gets a box and a number on the video, and a green **+1** appears when it crosses into the truck. The large number shows the total counted against the gate pass quantity.

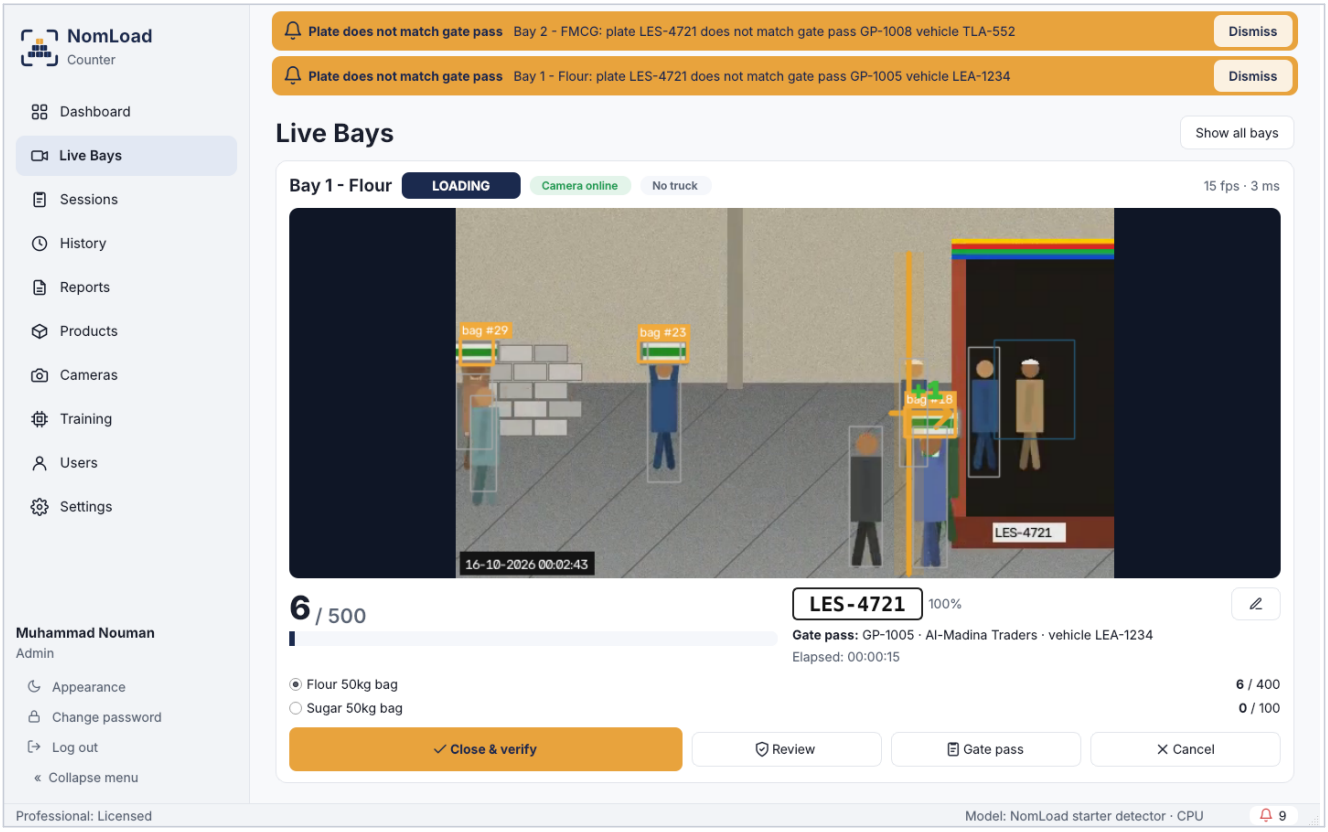


Figure 12: One bay enlarged (double-click the video). Counter, plate, gate pass and per-product counts

On screen	Meaning
Big number / expected	Units counted so far / quantity on the gate pass.
Progress bar	Blue while loading, green when complete, red if more than the gate pass.
Plate box	Number plate read by the camera with its confidence.
■ count(s) need review	Items the camera was not sure about. A supervisor checks them.
Product buttons	Select the product being loaded when two products look alike.
Camera online / Truck in bay	Camera and truck status.

7.3 Correcting the number plate

If the plate is wrong or empty, press the pencil button next to the plate, type the correct number and press **Save plate**. The correction is recorded in the audit log.

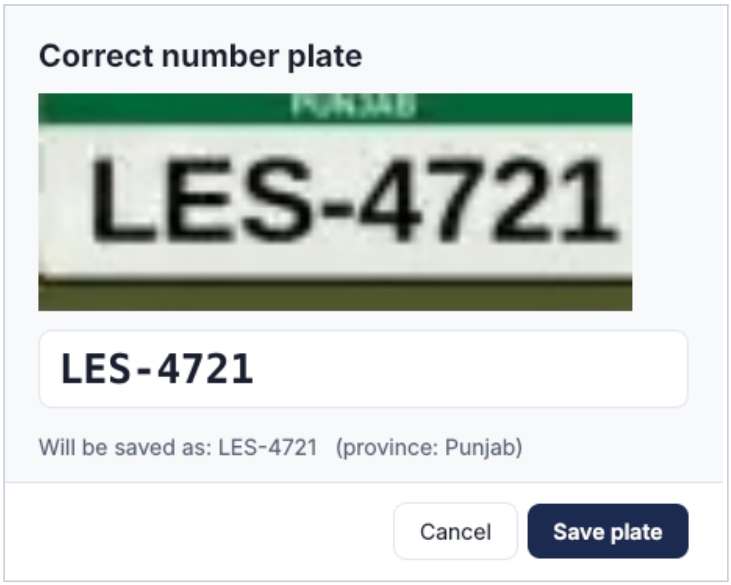


Figure 13: Correcting the plate

7.4 Reviewing doubtful counts

When an item was partly hidden, the count is marked for review instead of being added silently. Press **Review**, enter the supervisor PIN, look at each photo and press **Accept** or **Reject**.

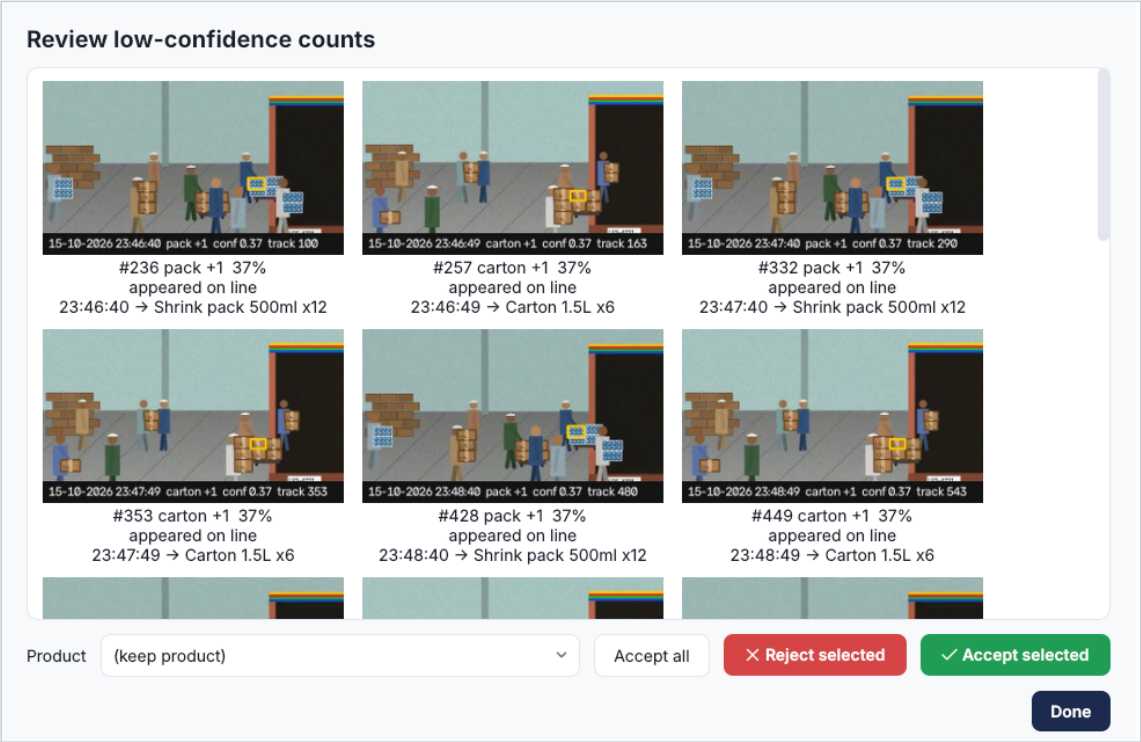


Figure 14: Supervisor review of doubtful counts with photos

7.5 Finishing: Close & verify

1. When loading is complete press **Close & verify**.
2. NomLoad compares every product with the gate pass, using the allowed tolerance.

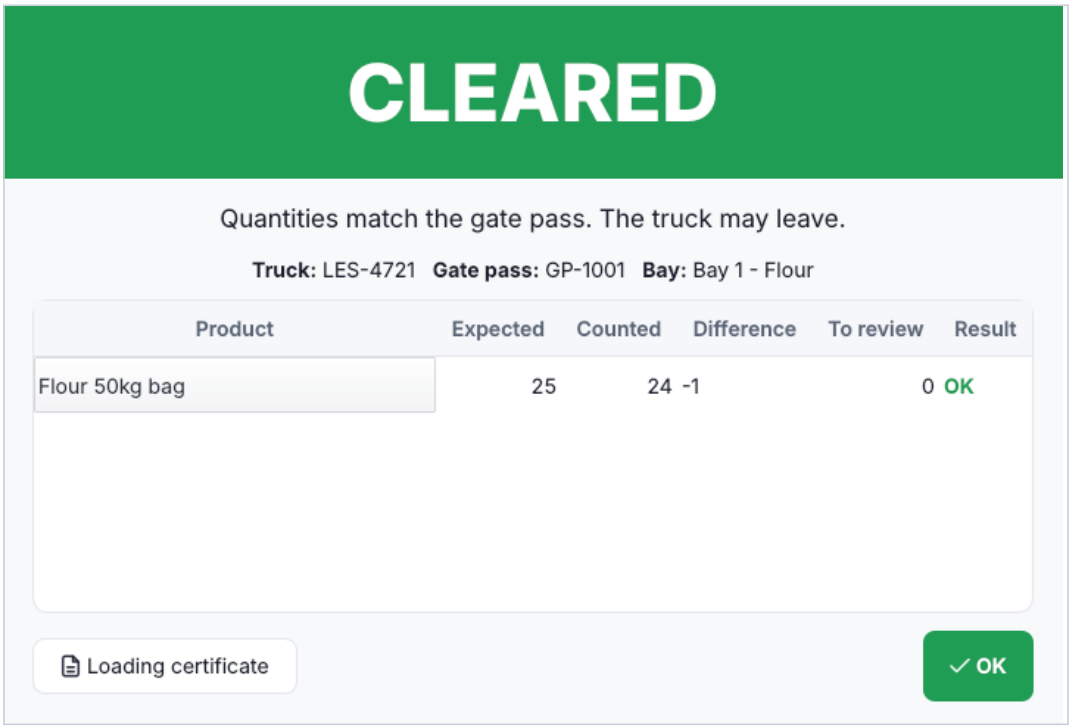


Figure 15: CLEARED: the truck may leave

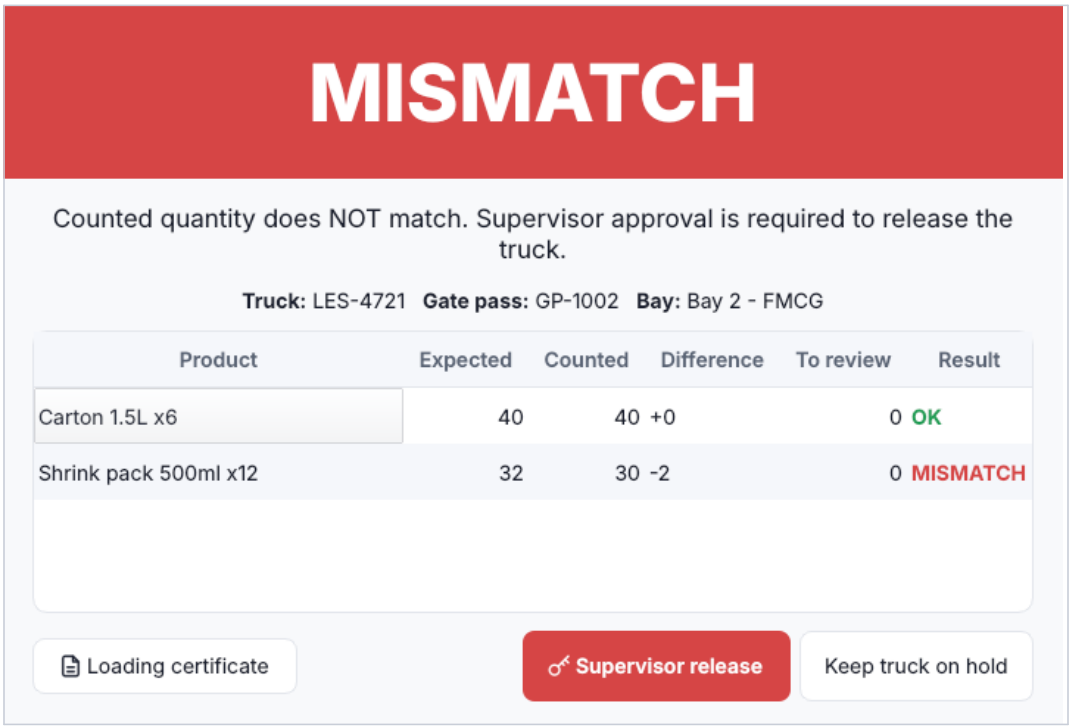


Figure 16: MISMATCH: the alarm sounds and the truck stays on hold

7.6 Releasing a mismatched truck

Only a supervisor can release a truck with a mismatch. Press **Supervisor release**, select the supervisor, enter the PIN and write the reason, for example *5 bags short, customer informed*.

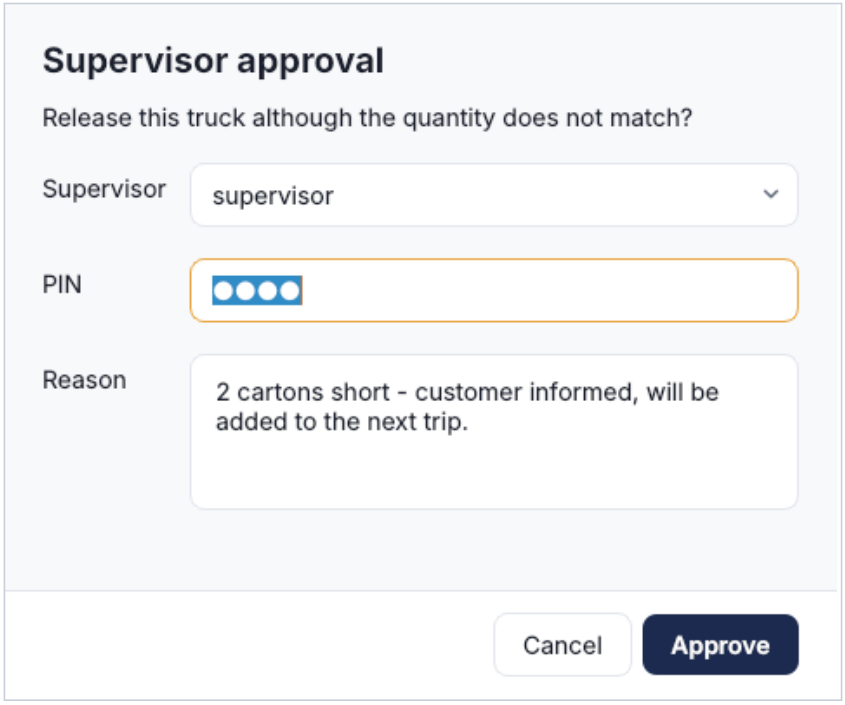


Figure 17: Supervisor approval with PIN and reason

If the truck leaves the bay while a session is open, NomLoad closes and checks the loading automatically (this can be turned off in Settings).

8. History and evidence

Open **History** to find any truck. Search by date, plate, gate pass number, product, bay or status, then double-click a row.

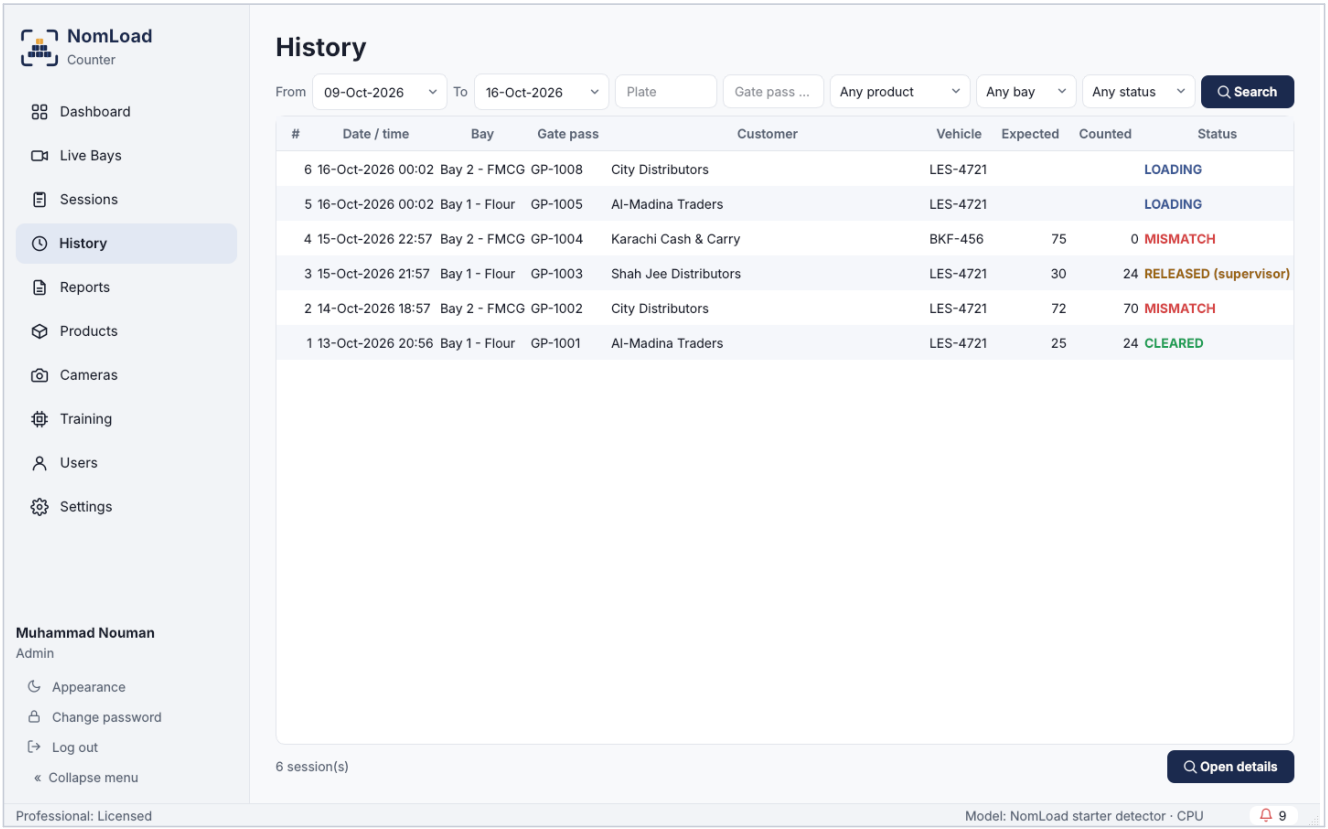


Figure 18: History search

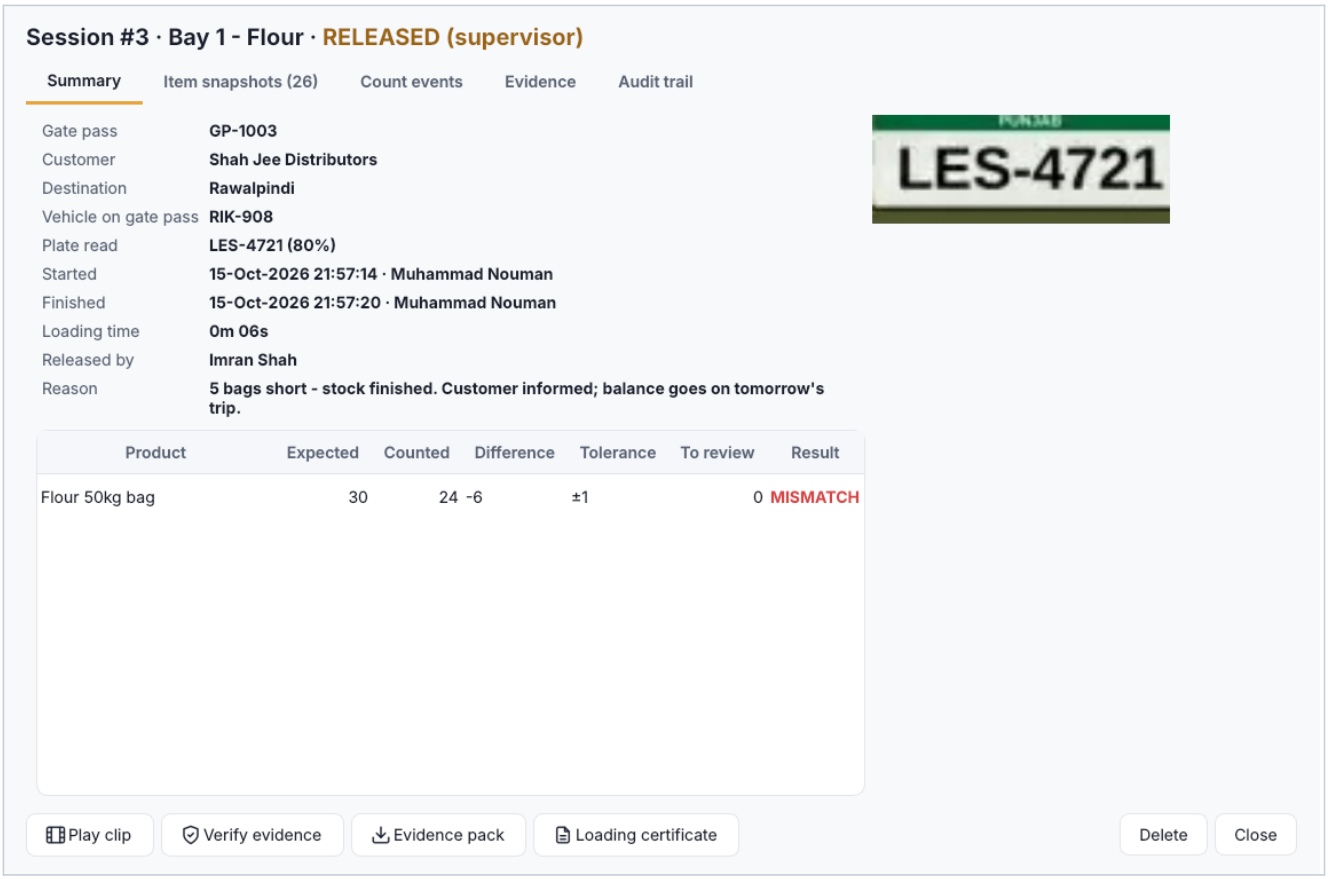


Figure 19: Session details: quantities, plate and who did what

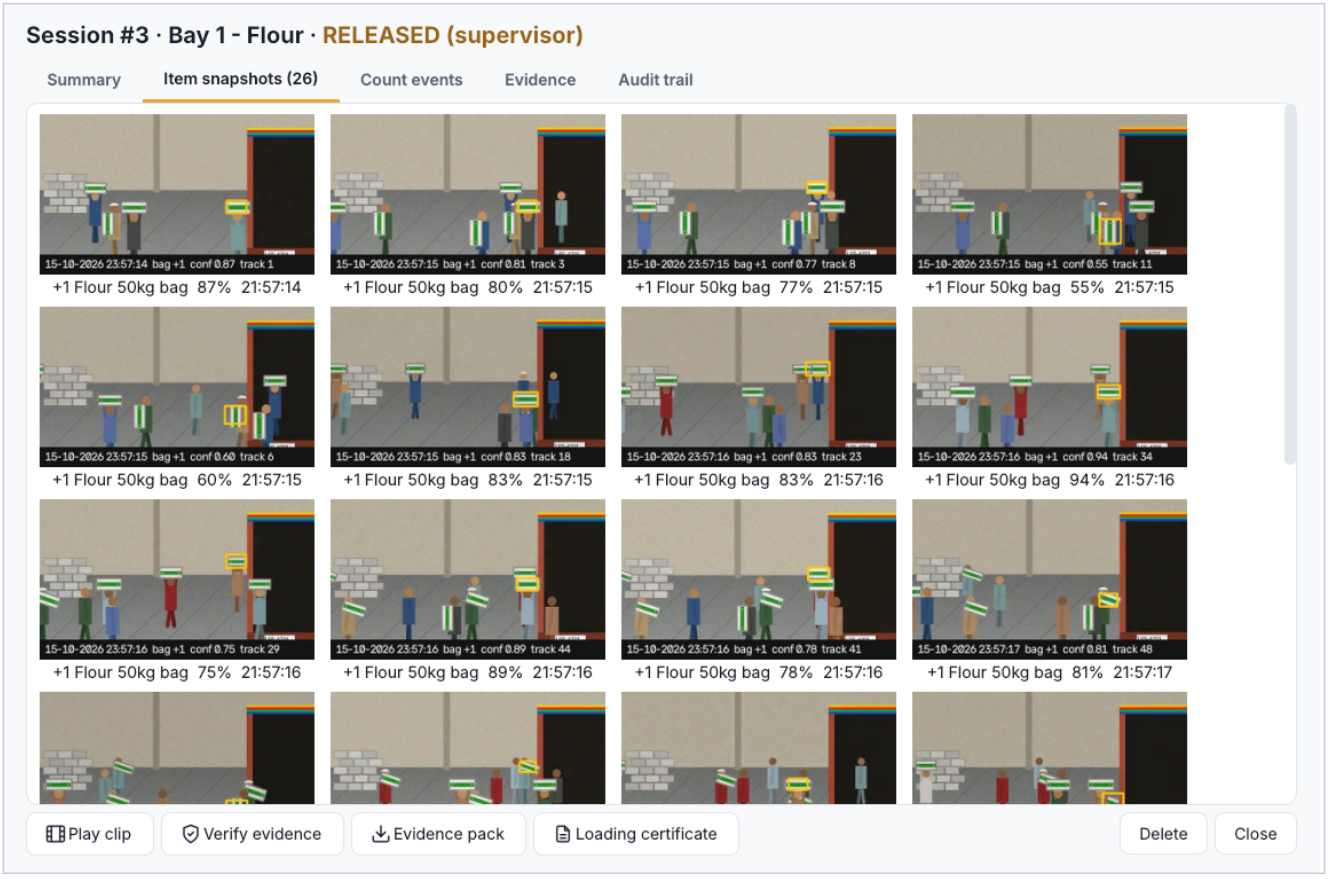


Figure 20: Photo of every counted item (double-click to enlarge)

8.1 Checking and sharing evidence

- **Play clip** opens the video of the whole loading.
- **Verify evidence** checks every file against its stored fingerprint. Any edited or missing file is shown in red.
- **Evidence pack** saves a ZIP file with the video, photos, plate image and a PDF summary to send to a distributor in a dispute.
- **Loading certificate** prints the PDF certificate for this truck.
- The **Audit trail** tab lists every action on this truck with user and time.

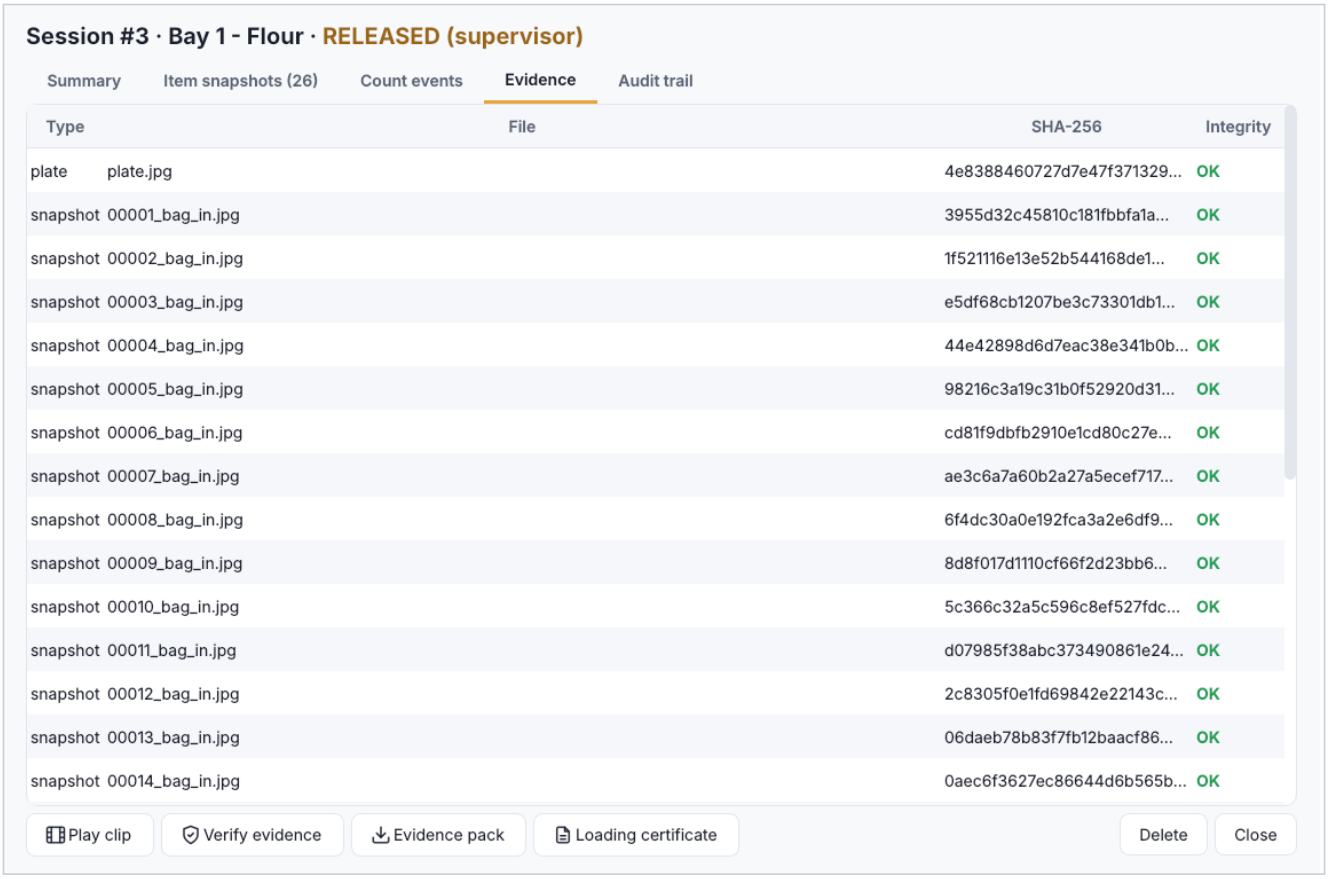


Figure 21: Evidence check: all files intact

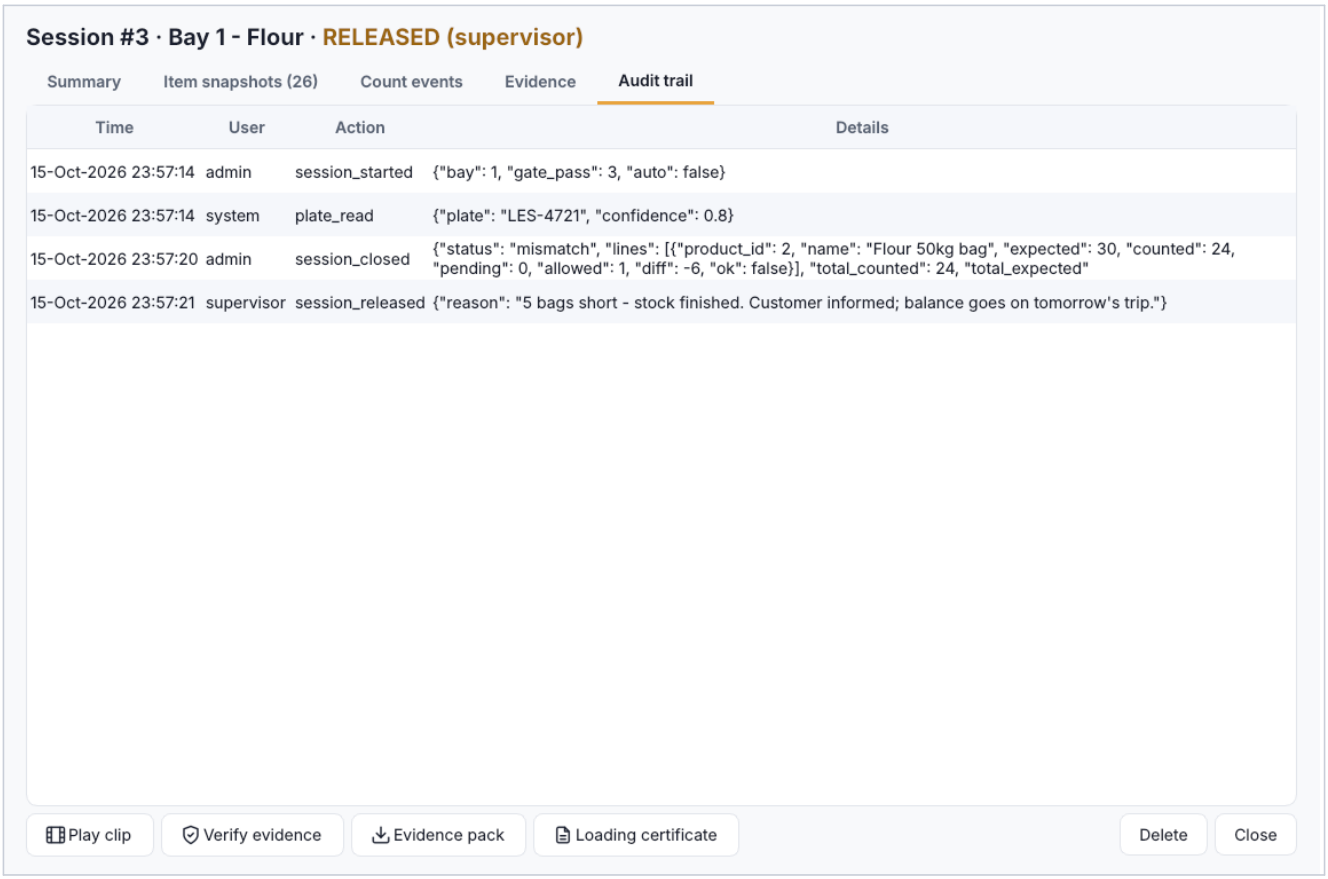


Figure 22: Audit trail of one truck

9. Reports

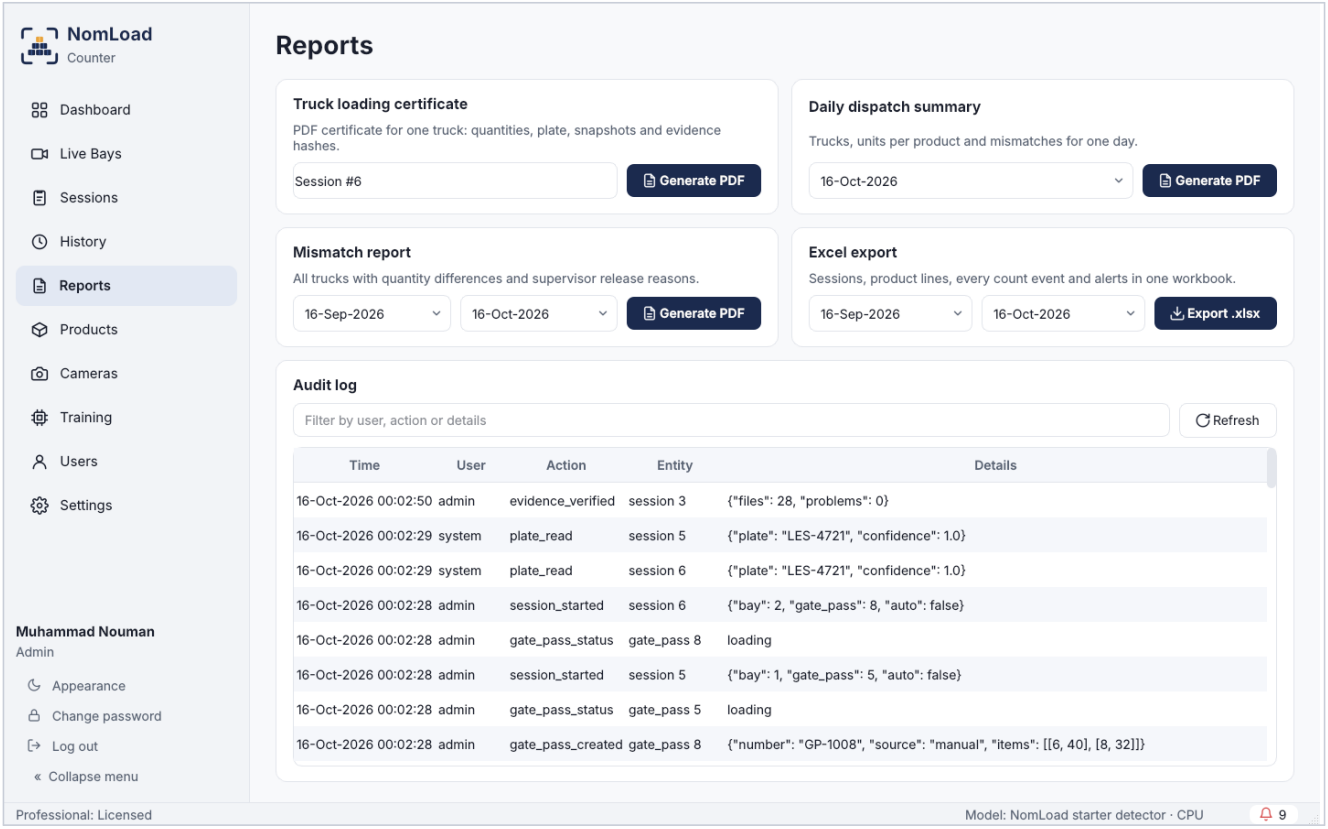
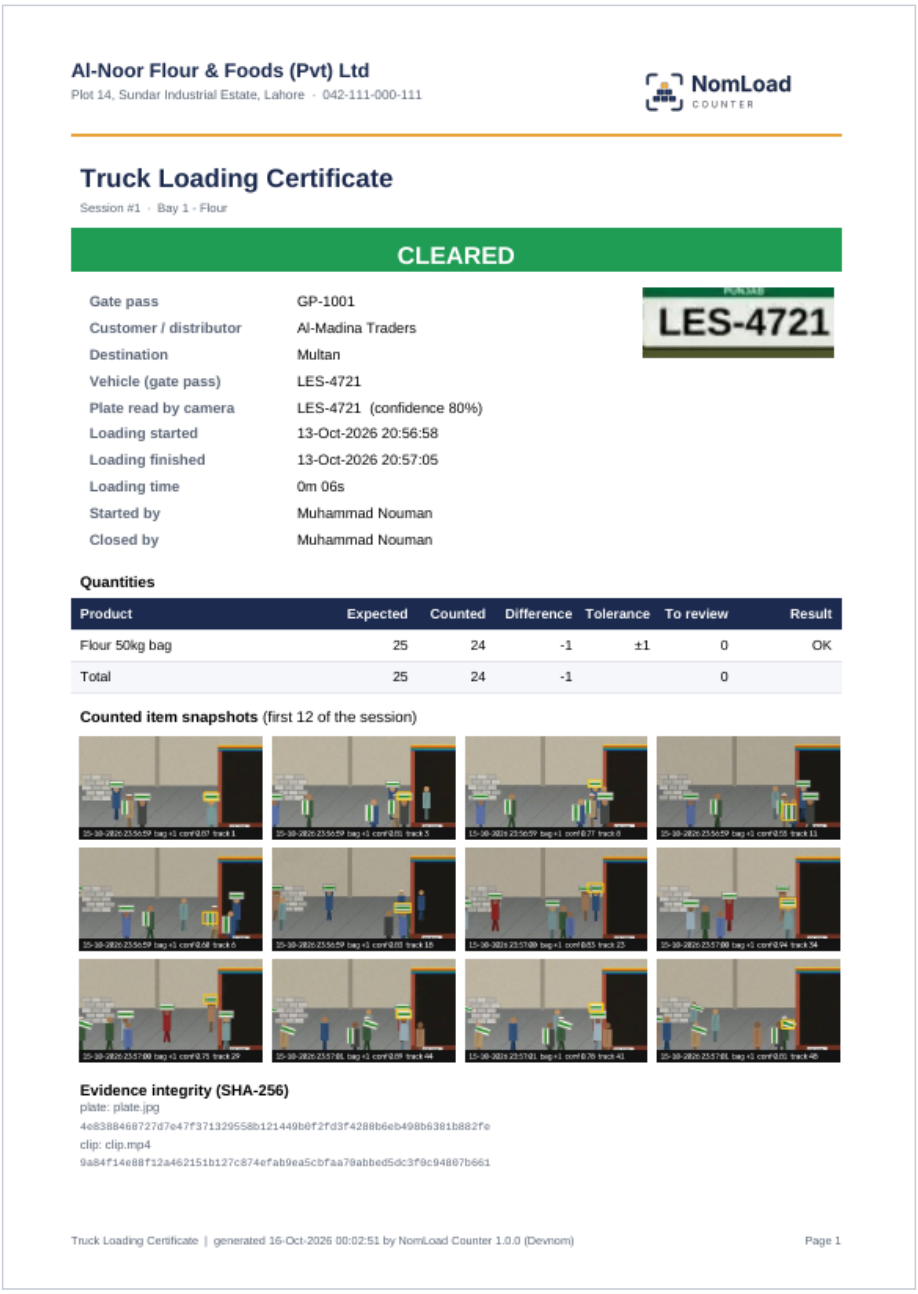


Figure 23: Reports page

Report	Contents
Truck loading certificate	One truck: quantities, plate, photos, evidence fingerprints and signature lines.
Daily dispatch summary	All trucks of one day, units per product and mismatches.
Mismatch report	Every truck with a difference and the supervisor's release reason.
Excel export	Sessions, products, every count event and alerts for a date range.
Audit log	Every action by every user. Records cannot be edited or deleted.



Al-Noor Flour & Foods (Pvt) Ltd
Plot 14, Sundar Industrial Estate, Lahore · 042-111-000-111

NomLoad COUNTER

Truck Loading Certificate

Session #1 · Bay 1 - Flour

CLEARED

Gate pass	GP-1001
Customer / distributor	Al-Madina Traders
Destination	Multan
Vehicle (gate pass)	LES-4721
Plate read by camera	LES-4721 (confidence 80%)
Loading started	13-Oct-2026 20:56:58
Loading finished	13-Oct-2026 20:57:05
Loading time	0m 06s
Started by	Muhammad Nouman
Closed by	Muhammad Nouman

LES-4721

Quantities

Product	Expected	Counted	Difference	Tolerance	To review	Result
Flour 50kg bag	25	24	-1	±1	0	OK
Total	25	24	-1		0	

Counted item snapshots (first 12 of the session)

Evidence integrity (SHA-256)

plate: plate.jpg
4e8388468727d7e47f371329558b121449b0f2fd3f4288b6eb498b6381b882fe
clip: clip.mp4
9a84f14e88f12a462151b127c874efab9ea5cbfaa70abbed5dc3f9c94807b661

Truck Loading Certificate | generated 16-Oct-2026 00:02:51 by NomLoad Counter 1.0.0 (Devnom) Page 1

Figure 24: Example loading certificate (PDF)



Al-Noor Flour & Foods (Pvt) Ltd
Plot 14, Sundar Industrial Estate, Lahore · 042-111-000-111

NomLoad COUNTER

Daily Dispatch Summary - Friday 16 October 2026

Trucks loaded	Units dispatched	Mismatches	Cleared first time
0	29	0	-

Units dispatched per product

Product	Units
Carton 1.5L x6	4
Flour 50kg bag	8
Shrink pack 500ml x12	17

Trucks

#	Time	Bay	Gate pass	Vehicle	Counted	Expected	Status
5	00:02	Bay 1 - Flour	GP-1005	LES-4721	-	-	LOADING
6	00:02	Bay 2 - FMCG	GP-1008	LES-4721	-	-	LOADING

Daily Dispatch Summary | generated 16-Oct-2026 00:02:51 by NomLoad Counter 1.0.0 (Devnom) Page 1

Figure 25: Example daily dispatch summary (PDF)

Your company name, address and logo are printed on every report. Set them in **Settings** → **Company**.

10. Training NomLoad for your products

Every factory's bags and cartons look different. For the best accuracy, train NomLoad with video from your own bay before going live. This takes about one to two hours of work.

10.1 Step 1: Collect frames

1. Export 10-30 minutes of normal loading from the NVR as an MP4 file.
2. Open **Training**, press **New dataset** and give it a name.
3. On **1 Collect frames** press **Choose video and extract**. NomLoad saves one picture every 15 frames (300 pictures by default).

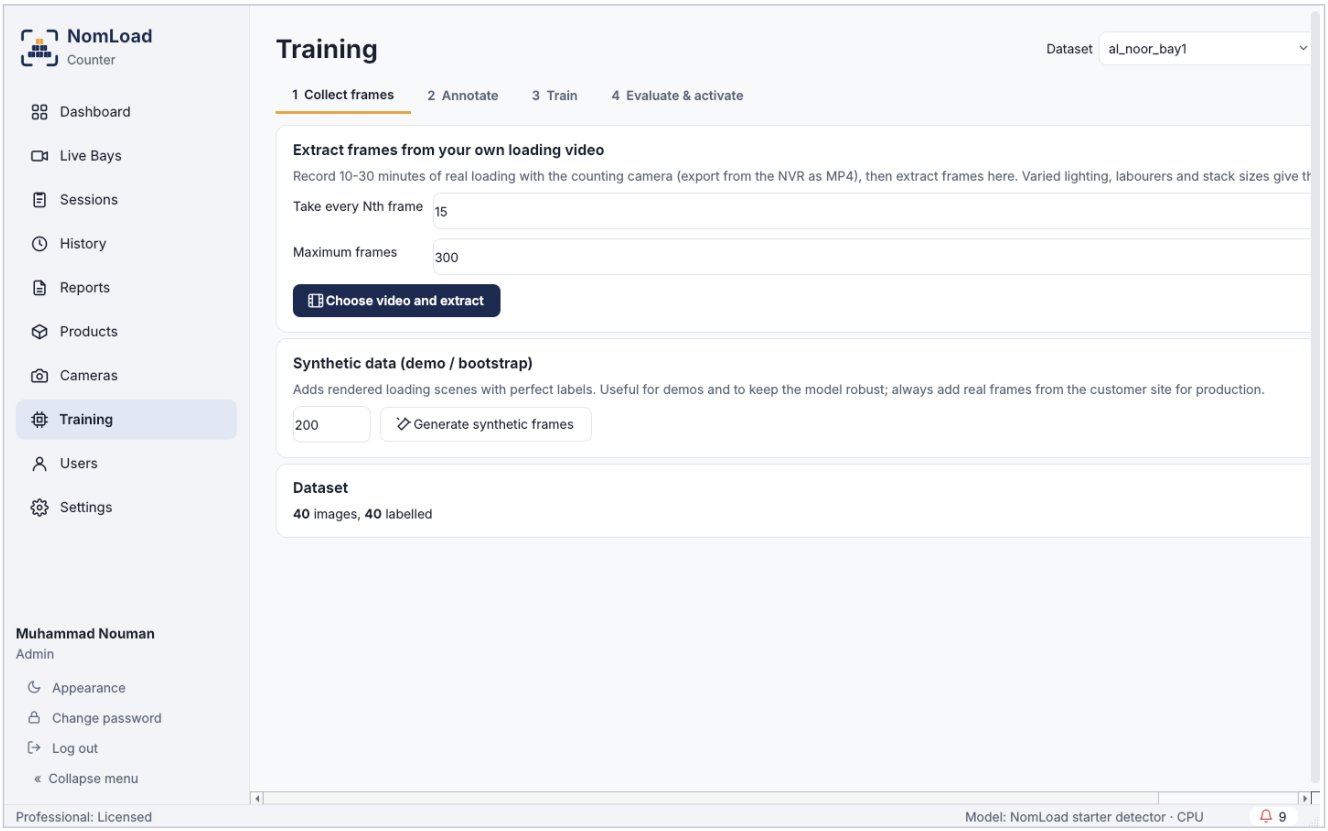


Figure 26: Collecting frames

10.2 Step 2: Annotate

1. Open **2 Annotate**. Press **Auto-label** so NomLoad draws the boxes it already recognises.
2. Correct the boxes: drag to draw a new box, select a box and press 1-9 to change its class, Delete to remove it.
3. Press **Save & next** (or PgDn) and continue until all pictures are done.

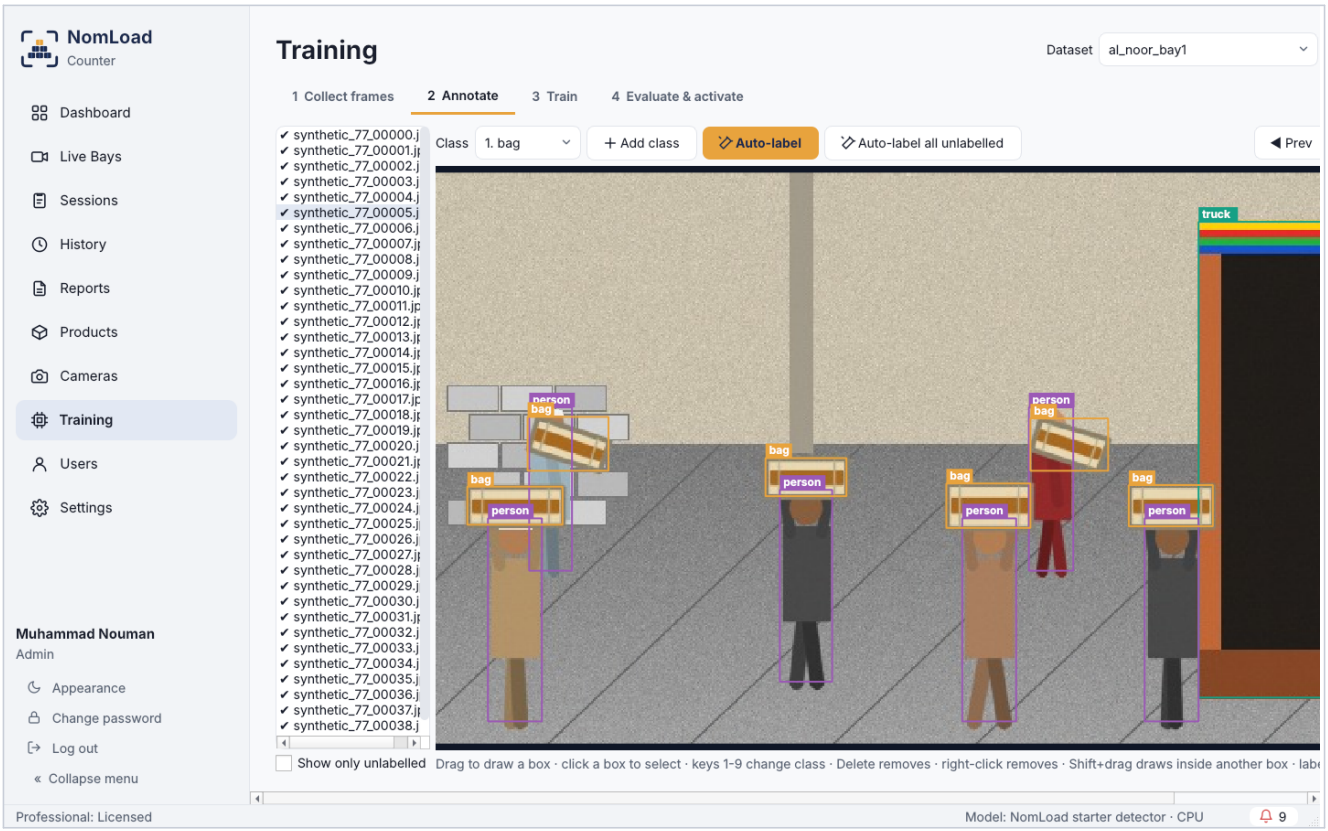


Figure 27: Annotation tool

10.3 Step 3: Train

On **3 Train** keep *Built-in trainer* and *Start from current model*, and press **Start training**. 1500 iterations take about 15 minutes. Counting continues while training runs.

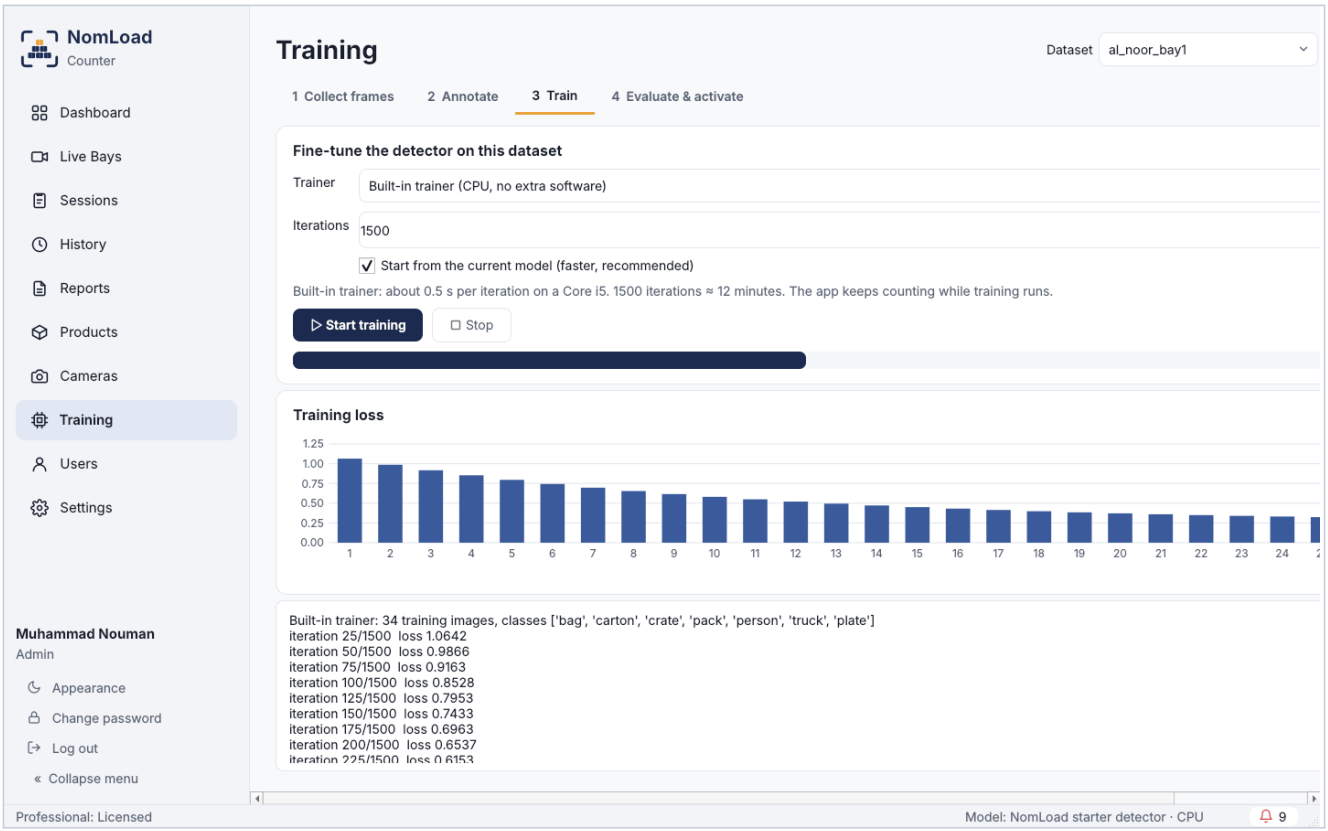


Figure 28: Training progress

10.4 Step 4: Activate

On **4 Evaluate & activate**, check precision and recall, then select the new model and press **Activate**. All bays switch to it immediately. The previous model stays in the list, so you can switch back.

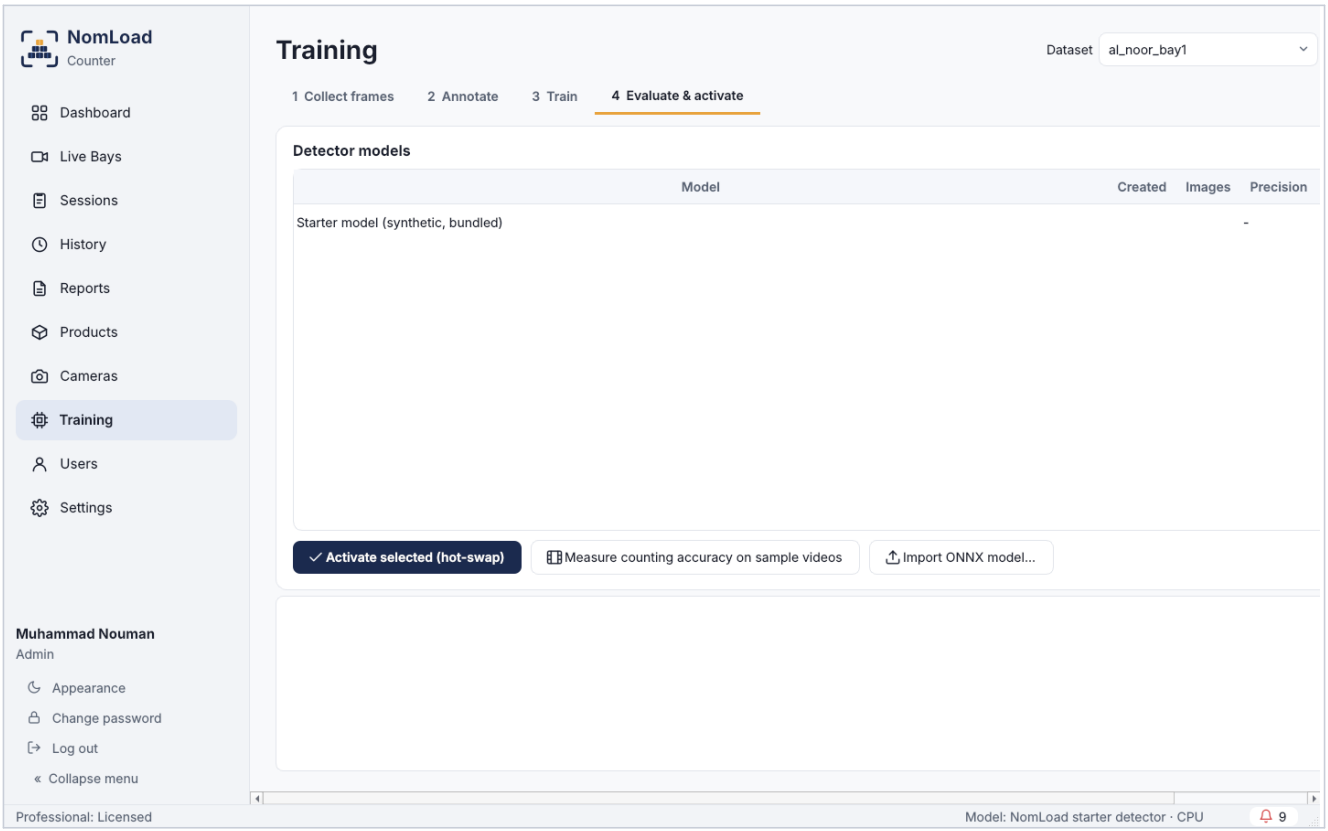


Figure 29: Models list

11. Users

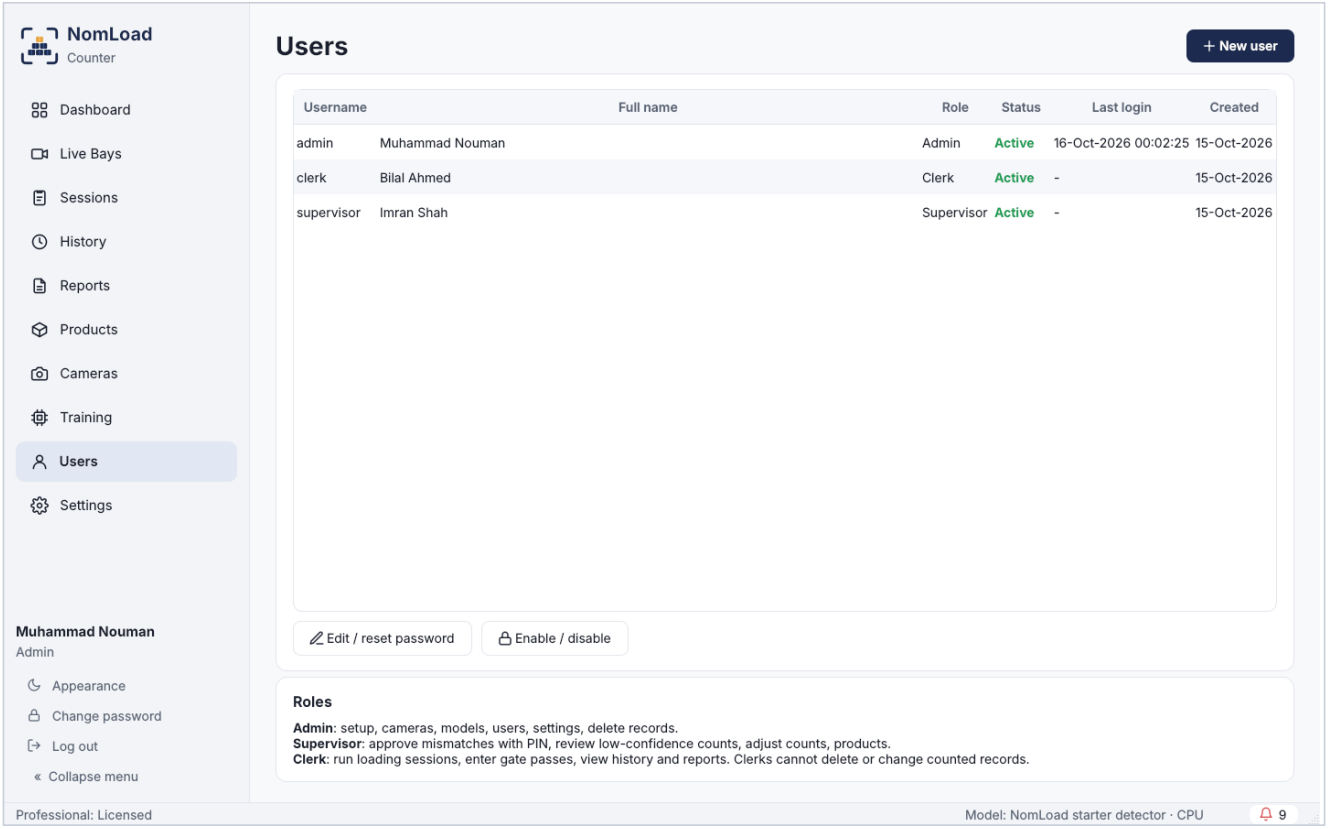


Figure 30: Users page

1. Press **New user**, enter username, full name and role.
2. Give a temporary password; the user must change it at first login.
3. Supervisors and Admins also need a **PIN** (4 or more digits) for approvals.
4. To stop someone using NomLoad, select the user and press **Enable / disable**. Users are never deleted, so the audit log stays complete.

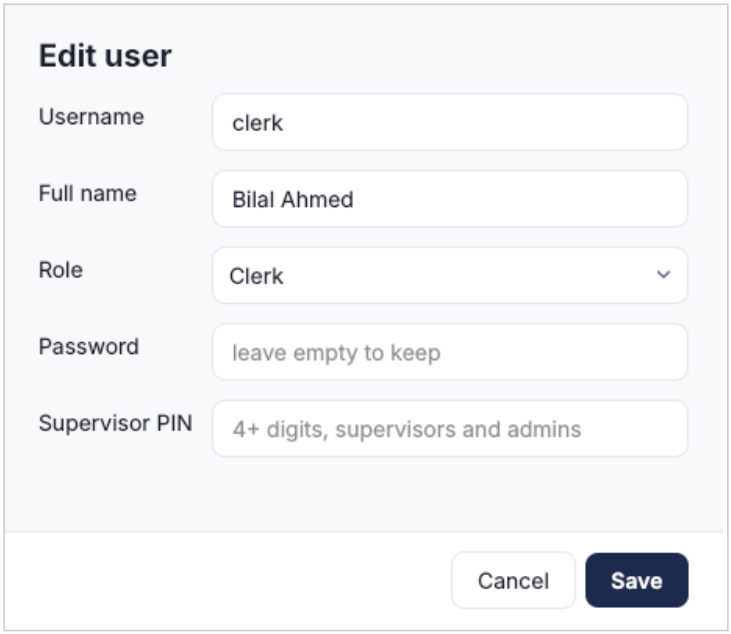


Figure 31: Editing a user

12. Settings

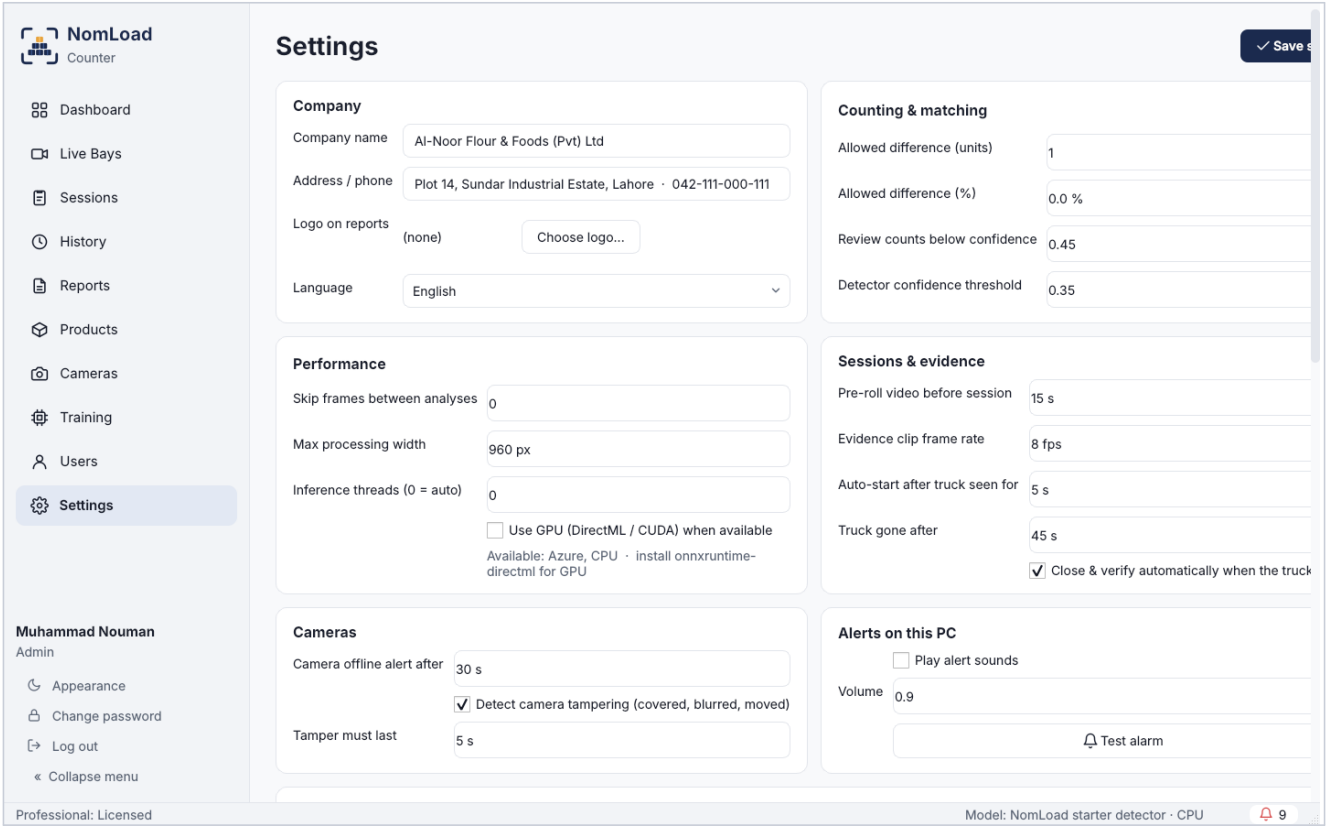


Figure 32: Settings

Section	Main settings
Company	Name, address and logo printed on reports; language.
Counting & matching	Allowed difference in units and %; confidence below which counts go to review.
Performance	Frames skipped between analyses, processing width, GPU. Raise *Skip frames* only if the PC is too slow.
Sessions & evidence	Video kept before a session, clip frame rate, automatic start and close.
Cameras	Offline alert time; tamper detection (covered, blurred or moved camera).
Alerts on this PC	Alarm sound and volume.
Remote alerts	Email, WhatsApp and SMS to the owner. Choose which events are sent. Messages are kept and sent when internet is available.
Licence	Status, PC request code, activate key, import revocation list.
Data	Data folder and database backup.

Press **Save settings** at the top after making changes.

12.1 Alerts

Alerts appear as a coloured bar at the top of the screen and in the bell list at the bottom-right.

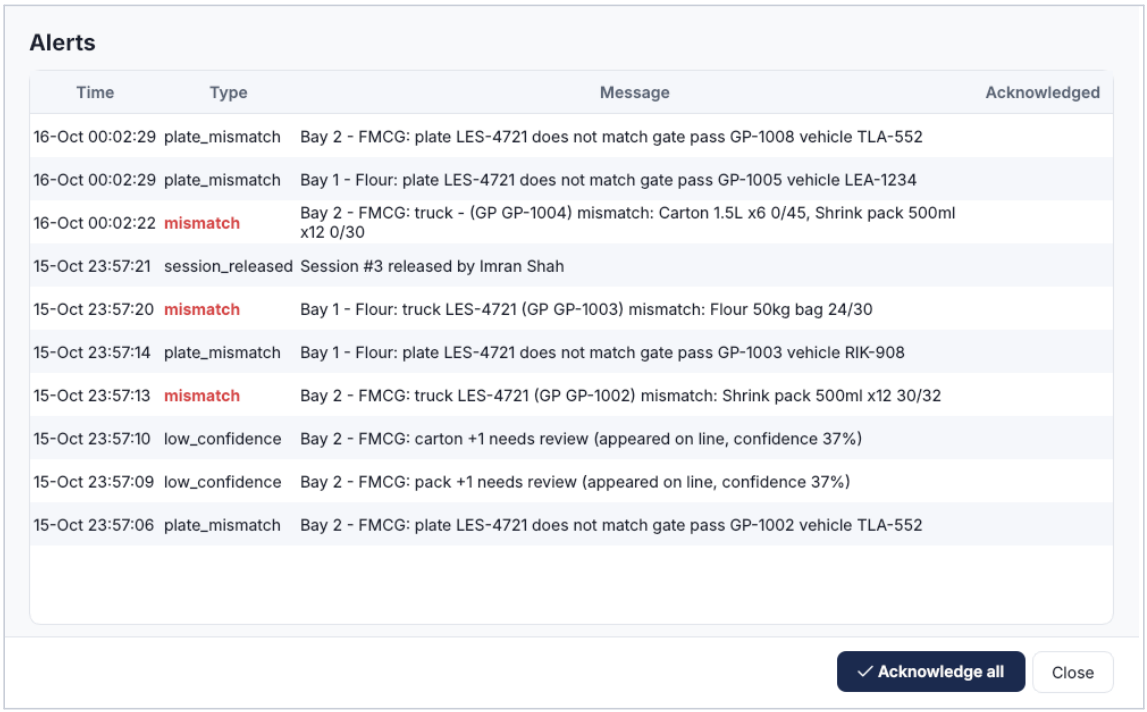


Figure 33: Alerts list

Alert	When
Quantity mismatch	A truck was closed with a difference (alarm sounds).
Camera offline	No video for longer than the set time.
Camera tampering	Camera covered, blurred, moved or pointed elsewhere.
Plate does not match	The plate read differs from the gate pass vehicle number.
Low-confidence count	An item needs supervisor review.

13. Appearance

Click **Appearance** at the bottom of the menu to switch between the light and dark look. Dark is easier on the eyes in a dim gate cabin.

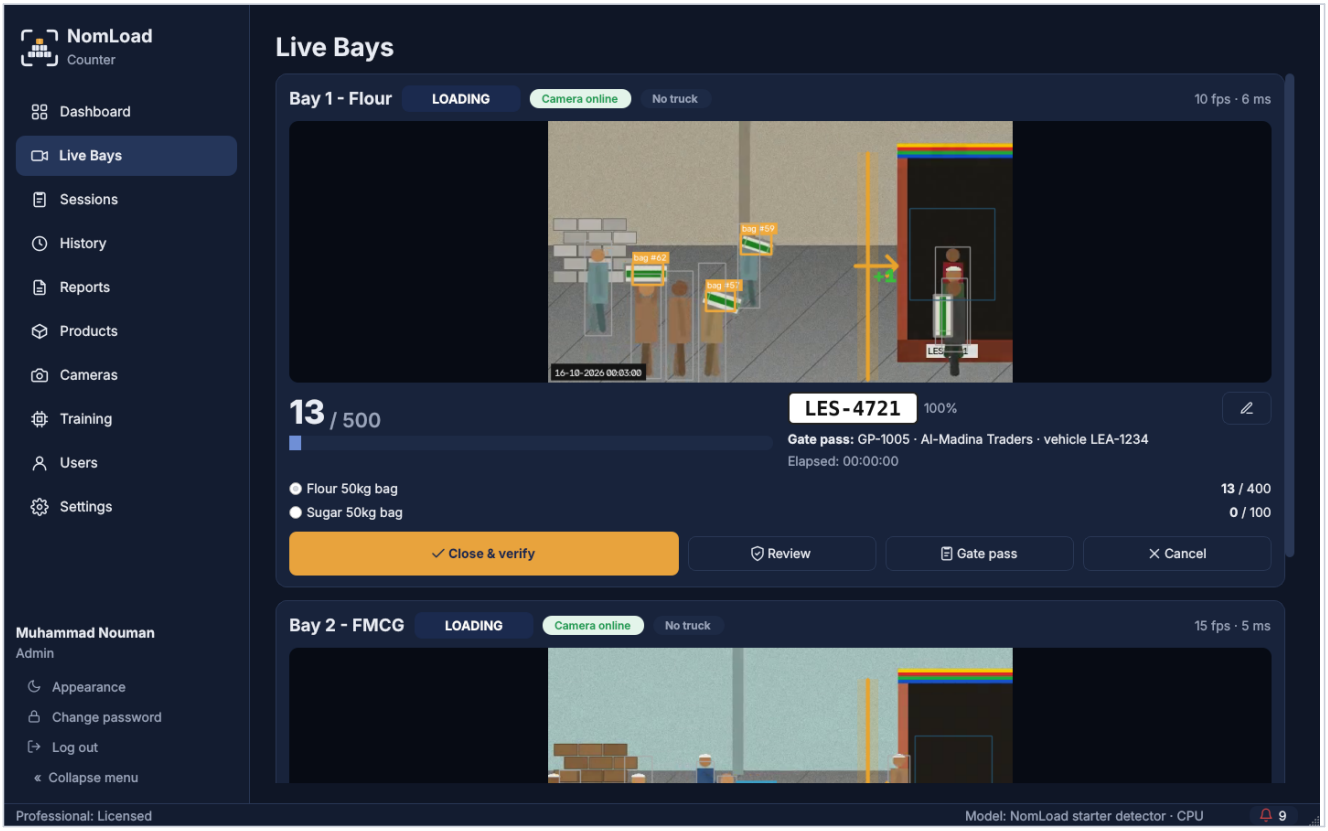


Figure 34: Dark appearance

14. Troubleshooting

Problem	What to do
Camera offline	Check the network cable and that the camera/NVR is powered. Open Cameras, select the camera, press Edit → Test connection. Check username, password and channel.
Picture appears but nothing is counted	Check the calibration: the arrow must point into the truck and the line must cross the path of the labourers.
Counts too high	Widen the line tolerance band; add ignore zones over the stock pile; train with your own video.
Counts too low	Mount the camera higher so items are not hidden behind labourers; use the main stream instead of sub-stream; set Skip frames to 0; train with your own video.
Plate not read	Draw the plate region in the calibration or add a plate camera; correct the plate manually with the pencil button.
Program slow	Use the camera sub-stream, lower Max processing width to 640, or raise Skip frames to 1.
Camera tampering alert	Check the camera is not covered or moved. If it was moved on purpose, calibrate it again.
Licence expired / not valid	Settings → Licence: activate a key. If the PC was replaced, send the PC request code to Devnom.
Forgot password	Ask an Admin to reset it under Users.

14.1 Getting help

The log file is in the data folder (**Settings** → **Data** → **Open folder**, then *logs*). Send it to Devnom support together with a description of the problem. To check that the installation is healthy, run **NomLoad.exe --selftest** from the installation folder.

15. Quick reference for the gate clerk

Step	Action
1	Truck arrives at the bay → **Live Bays** → **Start loading** → select the gate pass.
2	Check the plate. Wrong? Press the pencil and correct it.
3	If loading two similar products, select the product being loaded.
4	Watch the counter. ■ review messages: call the supervisor.
5	Loading finished → **Close & verify**.
6	**CLEARED** → truck may leave. Print the certificate if needed.
7	**MISMATCH** → truck stays. Call the supervisor for release or correction.

Daily checks: cameras show *Camera online* · camera lenses clean · no open alerts · end of day: print the **Daily dispatch summary**.